Linearised Polycrystals from a 2D System of Edge Dislocations

Silvio Fanzon

in collaboration with

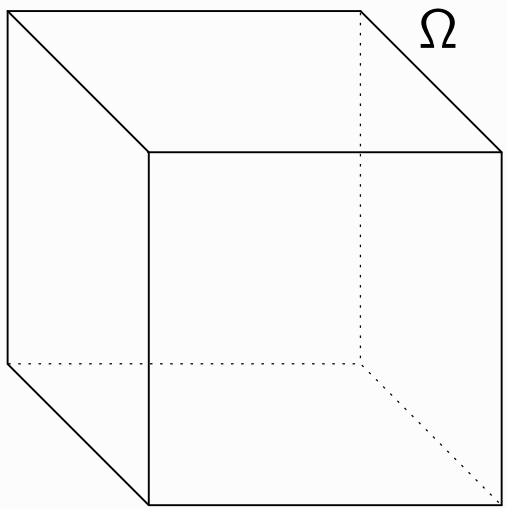
M. Palombaro and M. Ponsiglione

Graz, 31st January 2018

UNIVERSITY OF SUSSEX

Classical Elasticity: ideal crystals

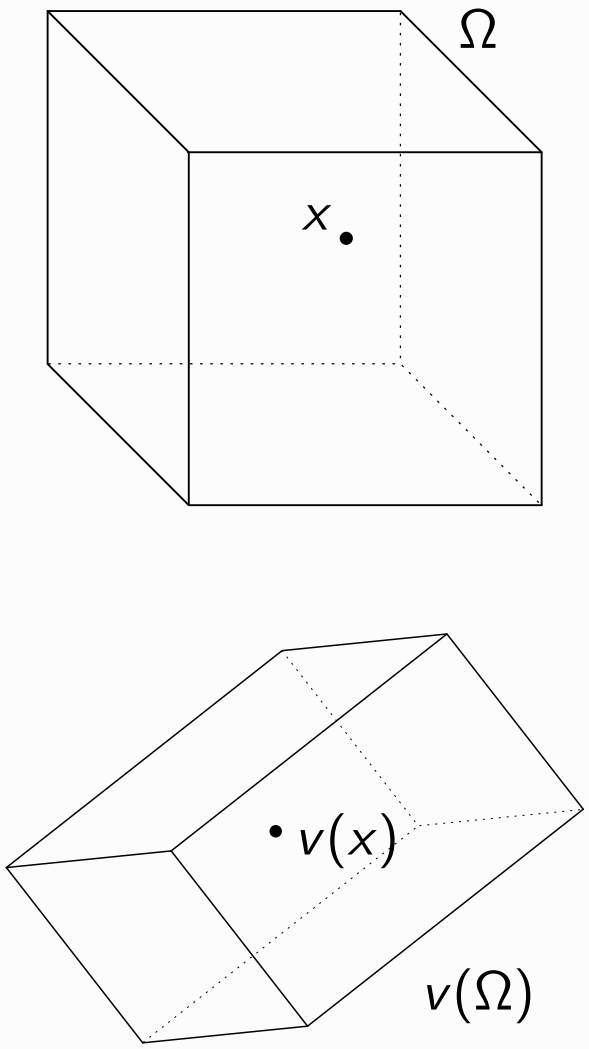
Reference configuration: $\Omega \subset \mathbb{R}^3$ open bounded



Classical Elasticity: ideal crystals

Reference configuration: $\Omega \subset \mathbb{R}^3$ open bounded

Deformations: regular maps $v\colon \Omega \to \mathbb{R}^3$

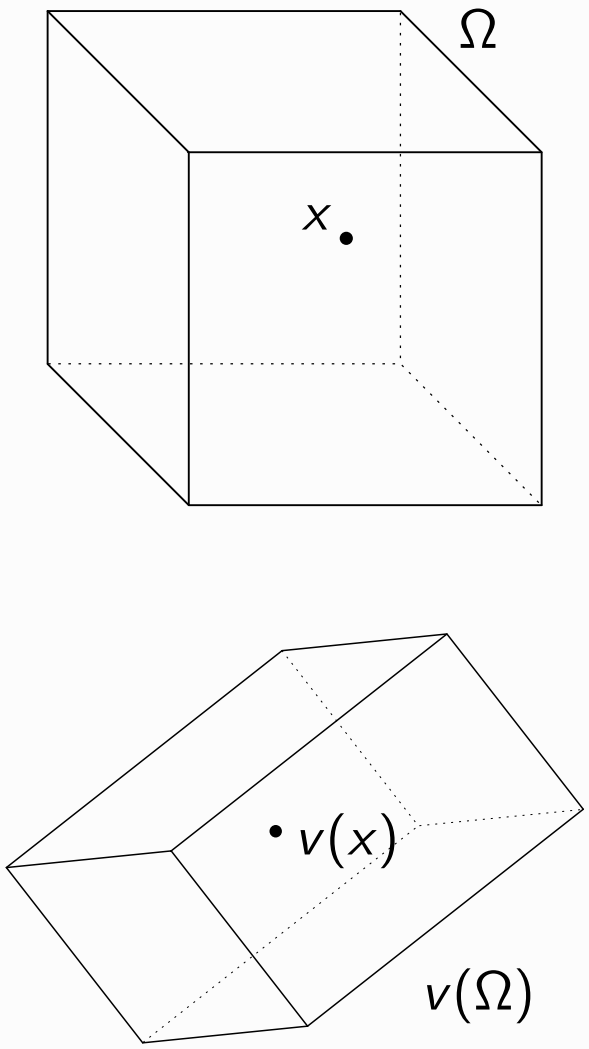


Classical Elasticity: ideal crystals

Reference configuration: $\Omega \subset \mathbb{R}^3$ open bounded

Deformations: regular maps $v : \Omega \to \mathbb{R}^3$

Deformation strain: $\beta := \nabla v : \Omega \to \mathbb{M}^{3\times 3}$



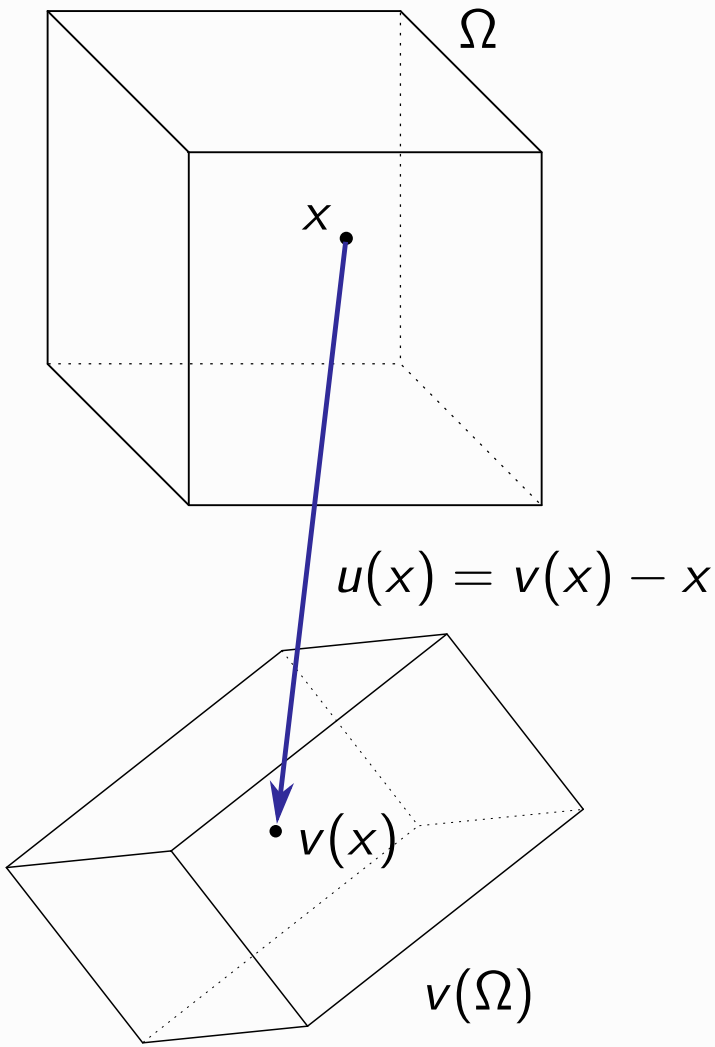
Classical Elasticity: ideal crystals

Reference configuration: $\Omega \subset \mathbb{R}^3$ open bounded

Deformations: regular maps $v\colon \Omega \to \mathbb{R}^3$

Deformation strain: $\beta := \nabla v\colon \Omega \to \mathbb{M}^{3\times 3}$

Displacement: map $u\colon \Omega \to \mathbb{R}^3$ s.t. $v = x + u$



Classical Elasticity: ideal crystals

Reference configuration: $\Omega \subset \mathbb{R}^3$ open bounded

Deformations: regular maps $v\colon \Omega \to \mathbb{R}^3$

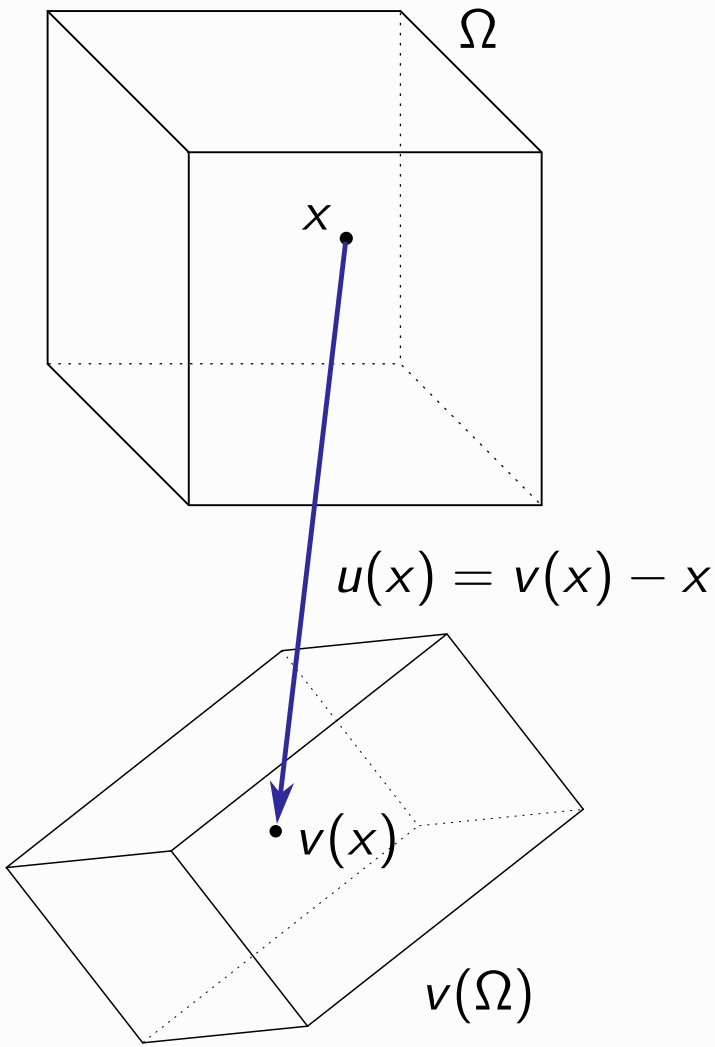
Deformation strain: $\beta := \nabla v\colon \Omega \to \mathbb{M}^{3\times 3}$

Displacement: map $u\colon \Omega \to \mathbb{R}^3$ s.t. $v = x + u$

Nonlinear Elasticity: the energy associated to a deformation strain β is

$$E(\beta) := \int_\Omega W(\beta)\, dx\,,$$

where $W(F) \sim \mathrm{dist}(F, SO(3))^2$ (so $W(I) = 0$).



Classical Elasticity: ideal crystals

Reference configuration: $\Omega \subset \mathbb{R}^3$ open bounded

Deformations: regular maps $v\colon \Omega \to \mathbb{R}^3$

Deformation strain: $\beta := \nabla v\colon \Omega \to \mathbb{M}^{3\times 3}$

Displacement: map $u\colon \Omega \to \mathbb{R}^3$ s.t. $v = x + u$

Nonlinear Elasticity: the energy associated to a deformation strain β is

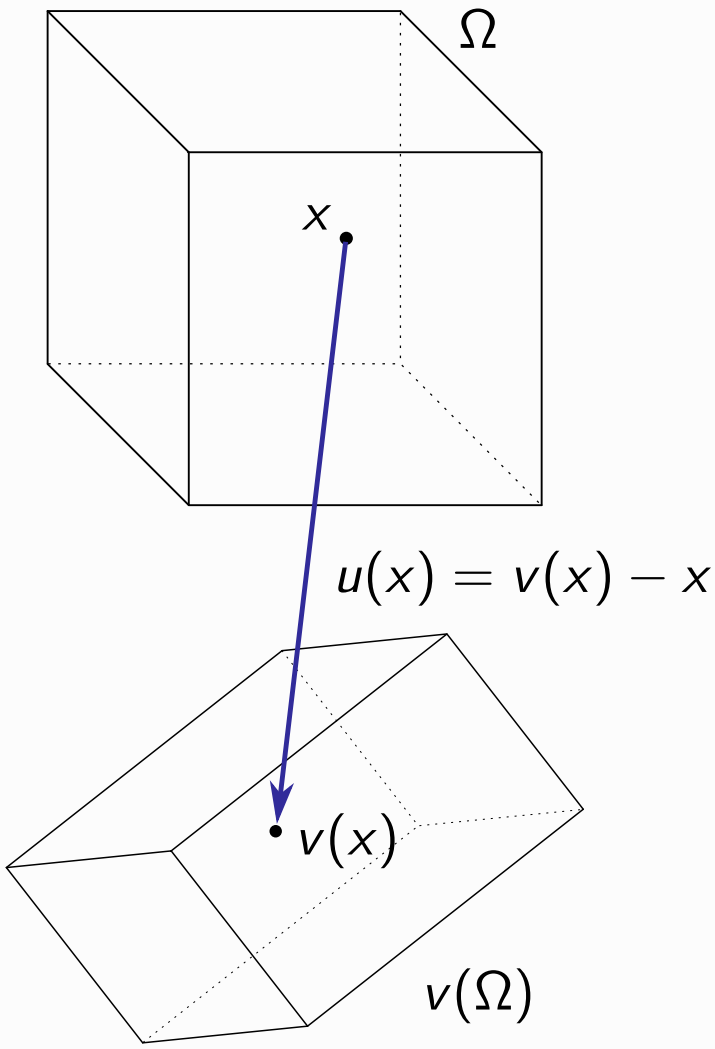
$$E(\beta) := \int_\Omega W(\beta)\, dx\,,$$

where $W(F) \sim \mathrm{dist}(F, SO(3))^2$ (so $W(I) = 0$).

Linear Elasticity: let $v = x + \varepsilon u$ with $\varepsilon \approx 0$. Then

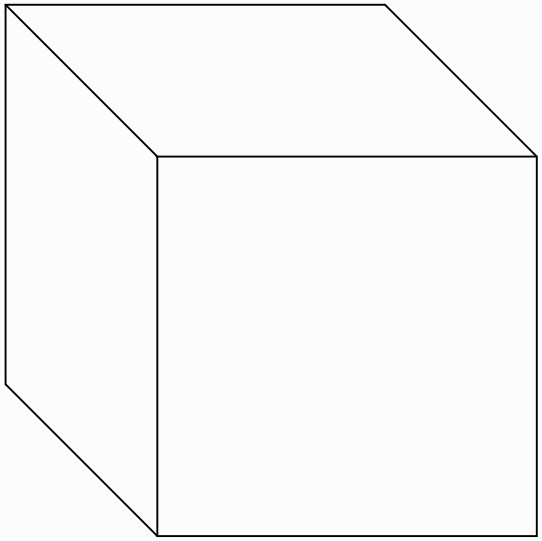
$$\lim_{\varepsilon\to 0} \frac{1}{\varepsilon^2} \int_\Omega W(\beta)\, dx = \frac{1}{2} \int_\Omega \mathbb{C}\nabla^{\mathrm{sym}} u : \nabla^{\mathrm{sym}} u\, dx\,,$$

where $\mathbb{C} = \partial^2 W(I)$ and $\nabla^{\mathrm{sym}} u := (\nabla u + \nabla u^T)/2$.



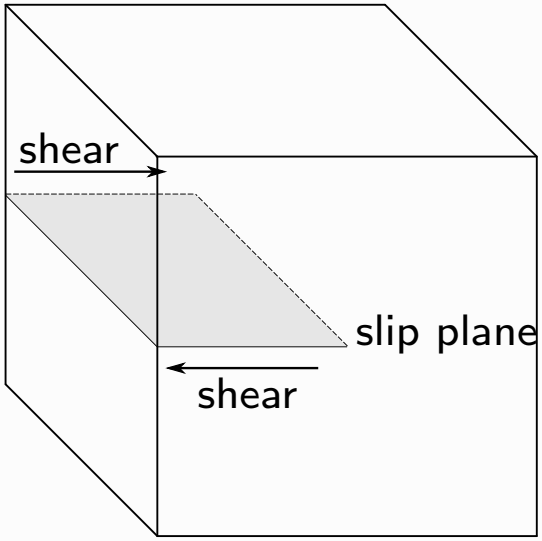
Edge dislocations

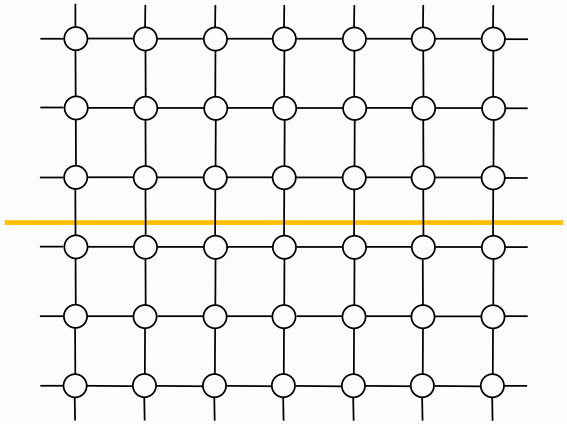
Dislocations: topological defects in the otherwise periodic structure of a crystal.



Edge dislocations

Dislocations: topological defects in the otherwise periodic structure of a crystal.

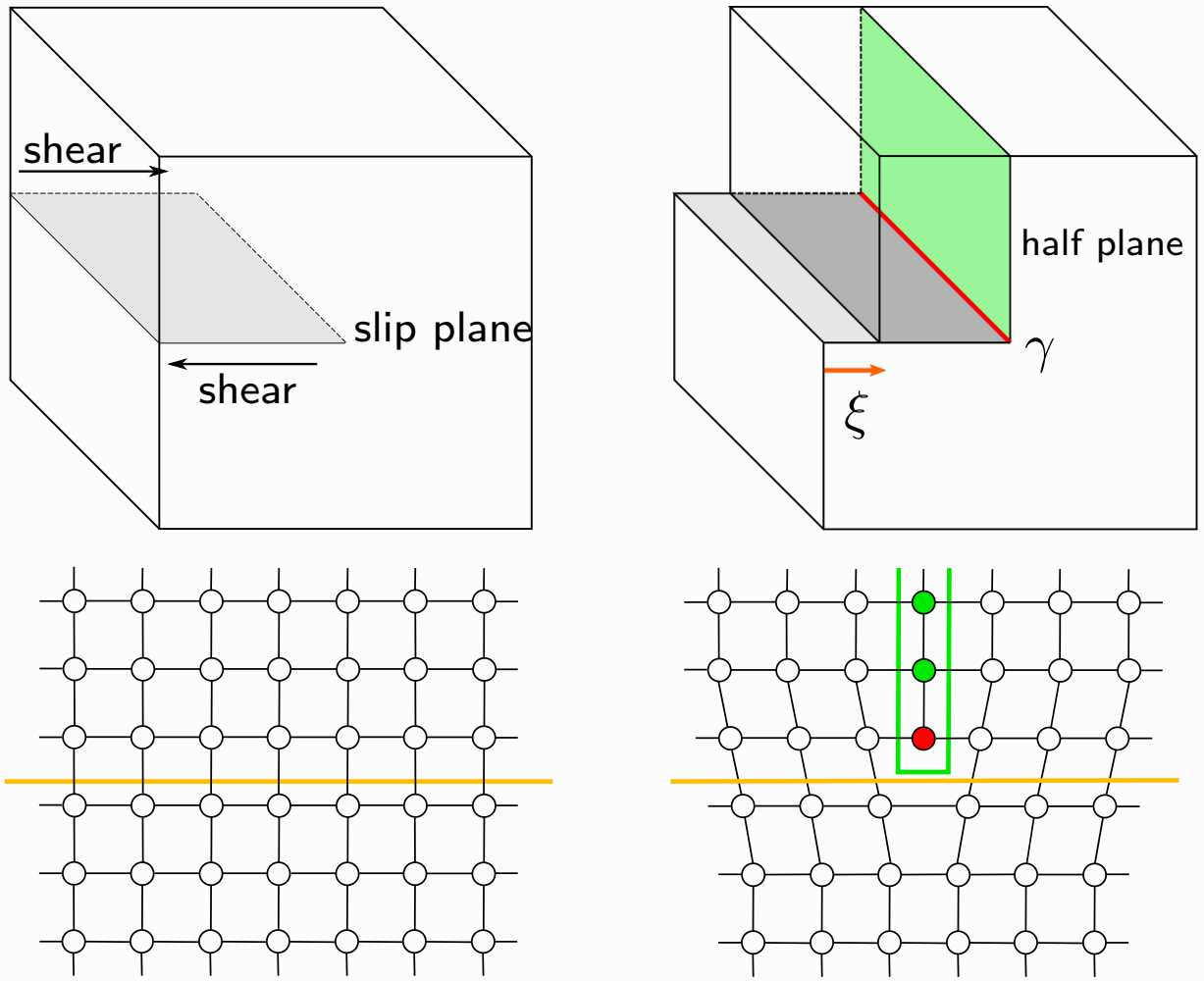




Edge dislocations

Dislocations: topological defects in the otherwise periodic structure of a crystal.

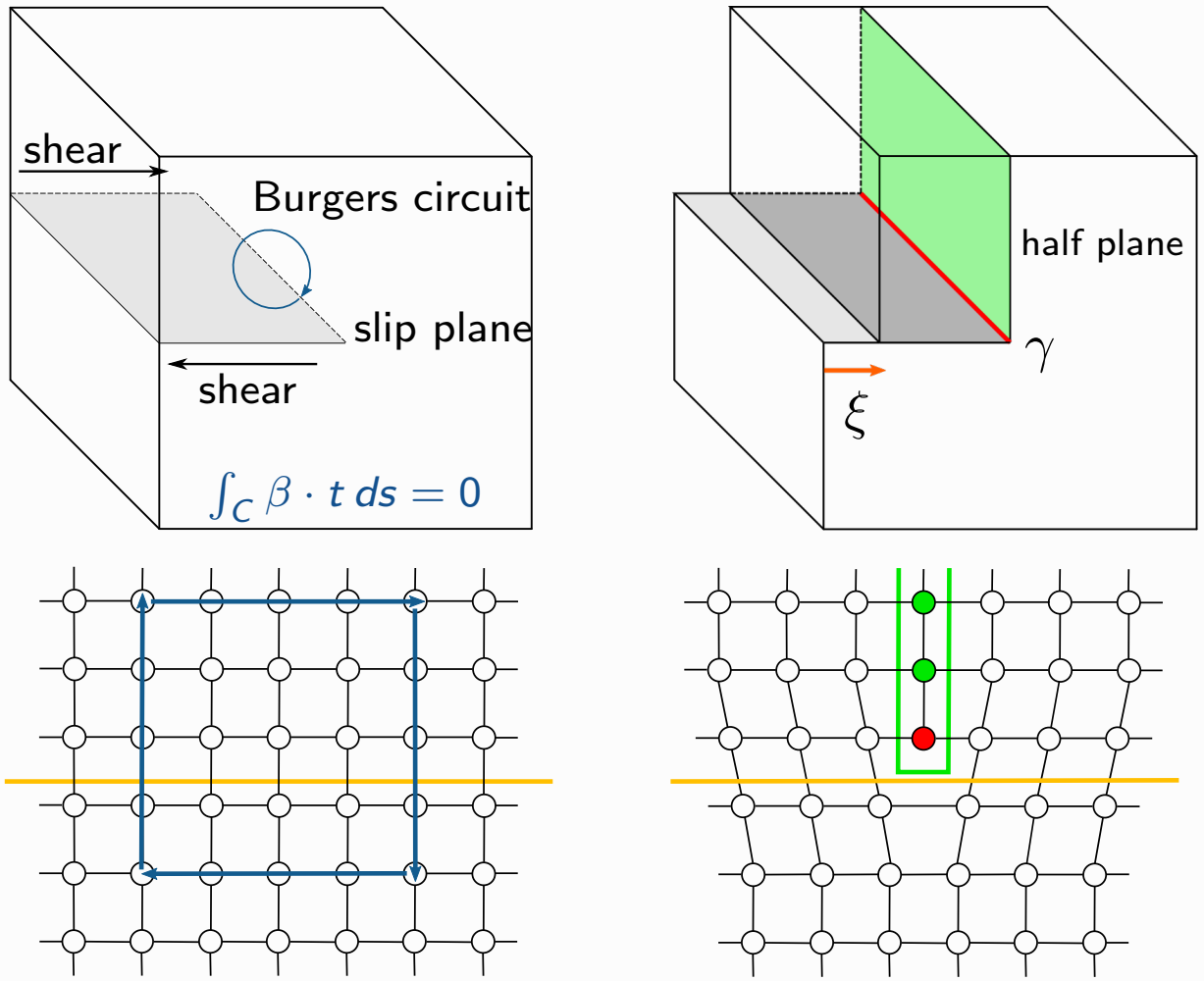
Edge dislocation: pair (γ, ξ) of dislocation line and Burgers vector, with $\xi \perp \gamma$.



Edge dislocations

Dislocations: topological defects in the otherwise periodic structure of a crystal.

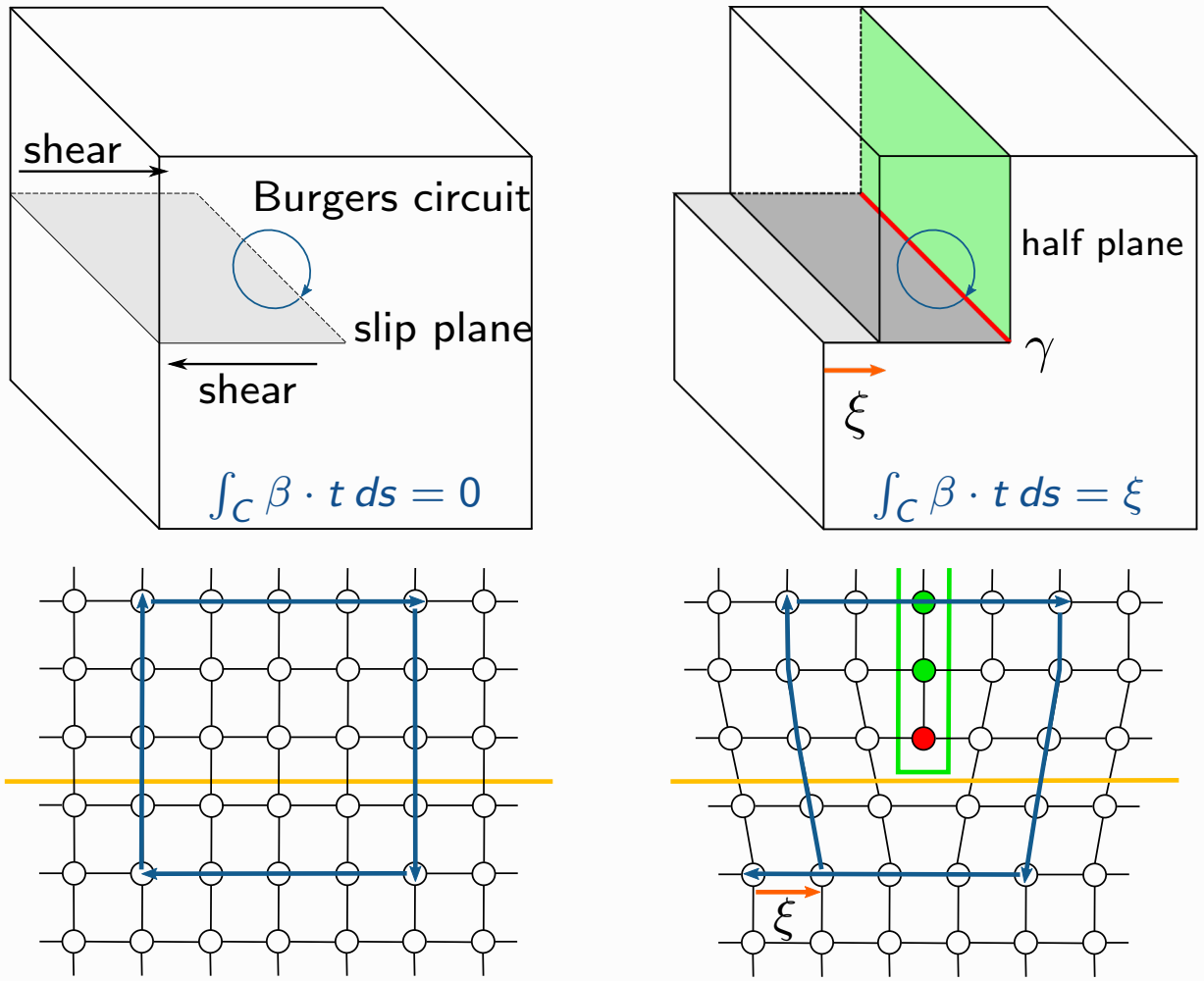
Edge dislocation: pair (γ, ξ) of dislocation line and Burgers vector, with $\xi \perp \gamma$.



Edge dislocations

Dislocations: topological defects in the otherwise periodic structure of a crystal.

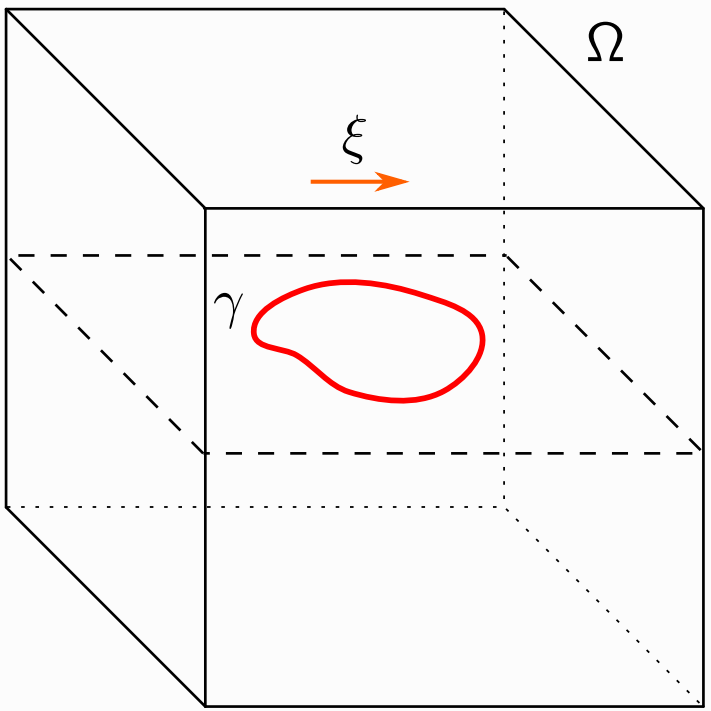
Edge dislocation: pair (γ, ξ) of dislocation line and Burgers vector, with $\xi \perp \gamma$.



Adding dislocations: the semi-discrete model

Dislocation lines: Lipschitz curves $\gamma \subset \Omega$ such that $\Omega \setminus \gamma$ is not simply connected

Burgers vector: $\xi \in \mathcal{S}$ set of slip directions



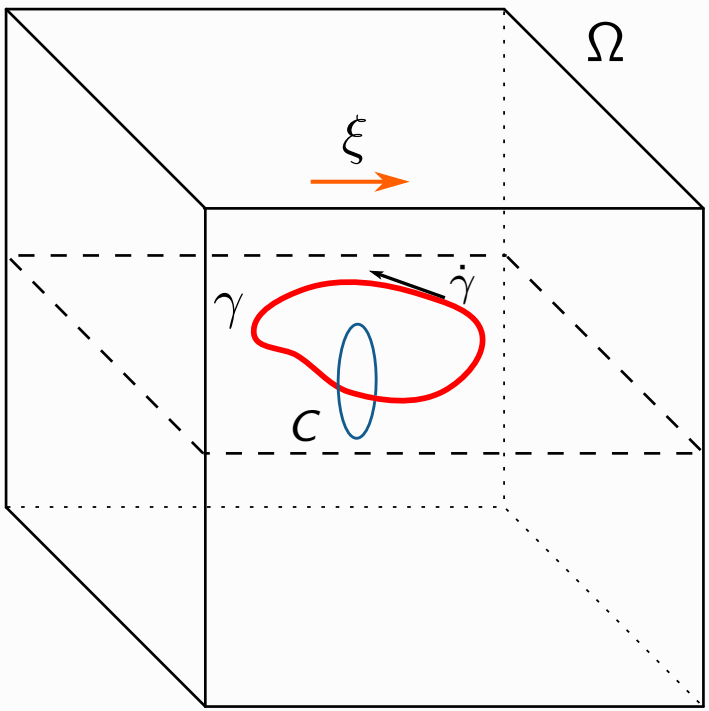
Adding dislocations: the semi-discrete model

Dislocation lines: Lipschitz curves $\gamma \subset \Omega$ such that $\Omega \setminus \gamma$ is not simply connected

Burgers vector: $\xi \in \mathcal{S}$ set of slip directions

Strain generating (γ, ξ): map $\beta \colon \Omega \to \mathbb{M}^{3\times 3}$ s.t.

$$\operatorname{Curl} \beta = -\xi \otimes \dot{\gamma}\, \mathcal{H}^1 \llcorner \gamma \implies \int_C \beta \cdot t \, d\mathcal{H}^1 = \xi \,.$$



Adding dislocations: the semi-discrete model

Dislocation lines: Lipschitz curves $\gamma \subset \Omega$ such that $\Omega \setminus \gamma$ is not simply connected

Burgers vector: $\xi \in \mathcal{S}$ set of slip directions

Strain generating (γ, ξ): map $\beta \colon \Omega \to \mathbb{M}^{3\times 3}$ s.t.

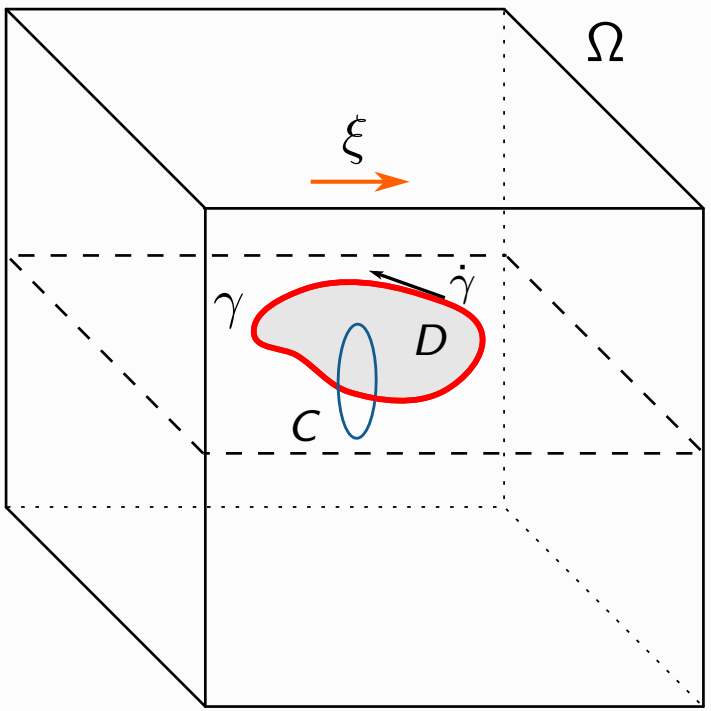
$$\operatorname{Curl} \beta = -\xi \otimes \dot{\gamma}\, \mathcal{H}^1 \llcorner \gamma \implies \int_C \beta \cdot t \, d\mathcal{H}^1 = \xi \,.$$

Geometric interpretation: if D encloses γ, there exists a deformation $v \in SBV(\Omega; \mathbb{R}^3)$ s.t.

$$Dv = \nabla v \, dx + \xi \otimes n \, \mathcal{H}^2 \llcorner D \,, \quad \beta = \nabla v \,.$$

In particular:

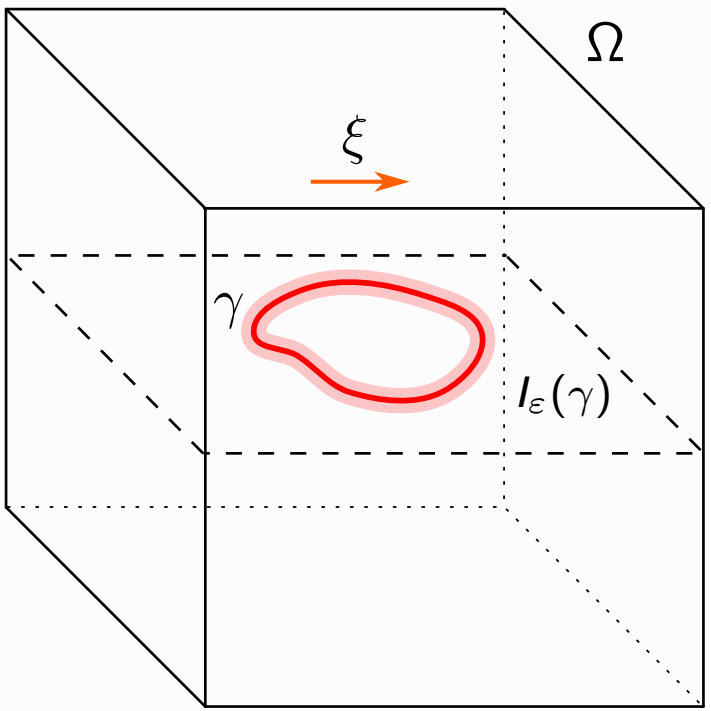
- $D =$ slip region,
- v has constant jump ξ across D,
- the absolutely continuous part of Dv is β.



Regularise the problem: Core Radius Approach

Let β generate (γ, ξ). Consider $\varepsilon > 0$ and

$$I_\varepsilon(\gamma) := \{x \in \mathbb{R}^3 : \ \mathrm{dist}(x, \gamma) < \varepsilon\}.$$



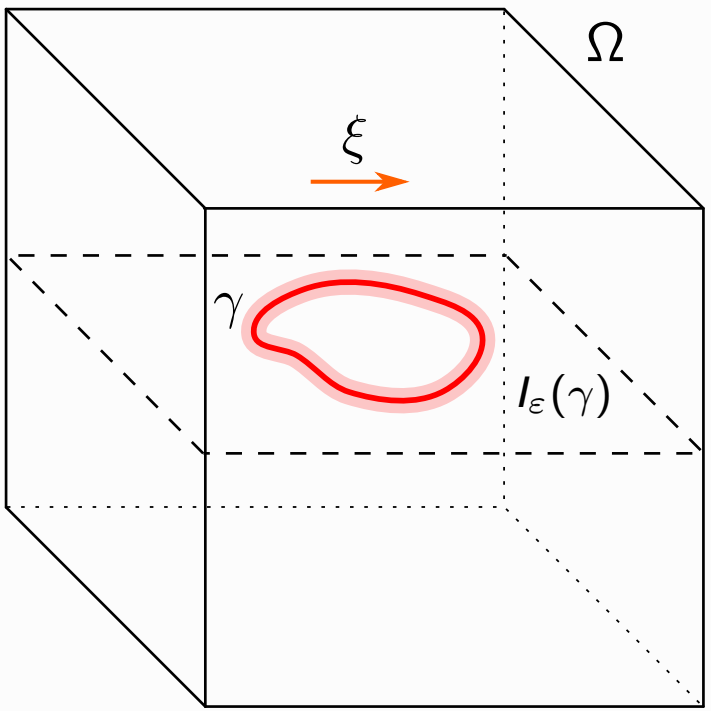
Regularise the problem: Core Radius Approach

Let β generate (γ, ξ). Consider $\varepsilon > 0$ and

$$I_\varepsilon(\gamma) := \{x \in \mathbb{R}^3 : \ \mathrm{dist}(x, \gamma) < \varepsilon\}.$$

Then we have

$$|\beta(x)| \sim \frac{1}{\mathrm{dist}(x, \gamma)} \ \text{ in } \ I_\varepsilon(\gamma) \implies \beta \notin L^2(I_\varepsilon(\gamma)).$$



Regularise the problem: Core Radius Approach

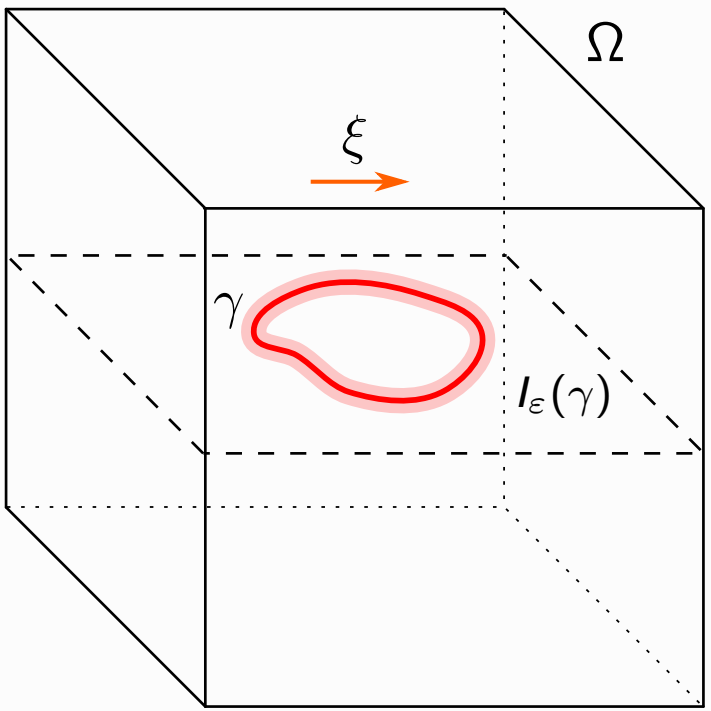
Let β generate (γ, ξ). Consider $\varepsilon > 0$ and

$$I_\varepsilon(\gamma) := \{x \in \mathbb{R}^3 : \ \mathrm{dist}(x, \gamma) < \varepsilon\}.$$

Then we have

$$|\beta(x)| \sim \frac{1}{\mathrm{dist}(x, \gamma)} \ \text{ in } \ I_\varepsilon(\gamma) \implies \beta \notin L^2(I_\varepsilon(\gamma)).$$

CRA: new ref. conf. $\Omega_\varepsilon(\gamma) := \Omega \setminus I_\varepsilon(\gamma)$.



Regularise the problem: Core Radius Approach

Let β generate (γ, ξ). Consider $\varepsilon > 0$ and

$$I_\varepsilon(\gamma) := \{x \in \mathbb{R}^3 : \ \mathrm{dist}(x, \gamma) < \varepsilon\}\,.$$

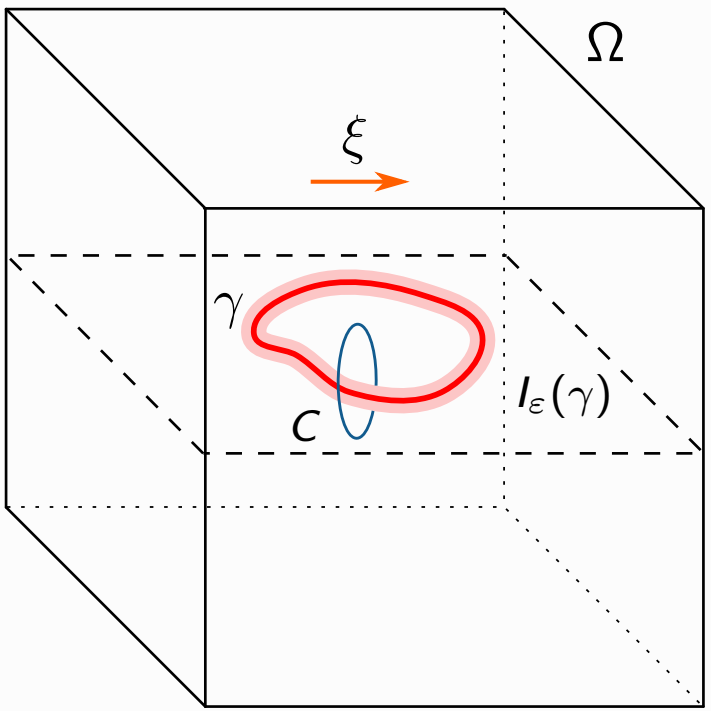
Then we have

$$|\beta(x)| \sim \frac{1}{\mathrm{dist}(x,\gamma)} \ \text{ in } \ I_\varepsilon(\gamma) \implies \beta \notin L^2(I_\varepsilon(\gamma)).$$

CRA: new ref. conf. $\Omega_\varepsilon(\gamma) := \Omega \setminus I_\varepsilon(\gamma)$.

New Strains: maps $\beta \in L^2(\Omega_\varepsilon(\gamma); \mathbb{M}^{3\times 3})$ s.t.

$$\mathrm{Curl}\,\beta \llcorner \Omega_\varepsilon(\gamma) = 0\,, \qquad \int_C \beta \cdot t \, d\mathcal{H}^1 = \xi\,.$$



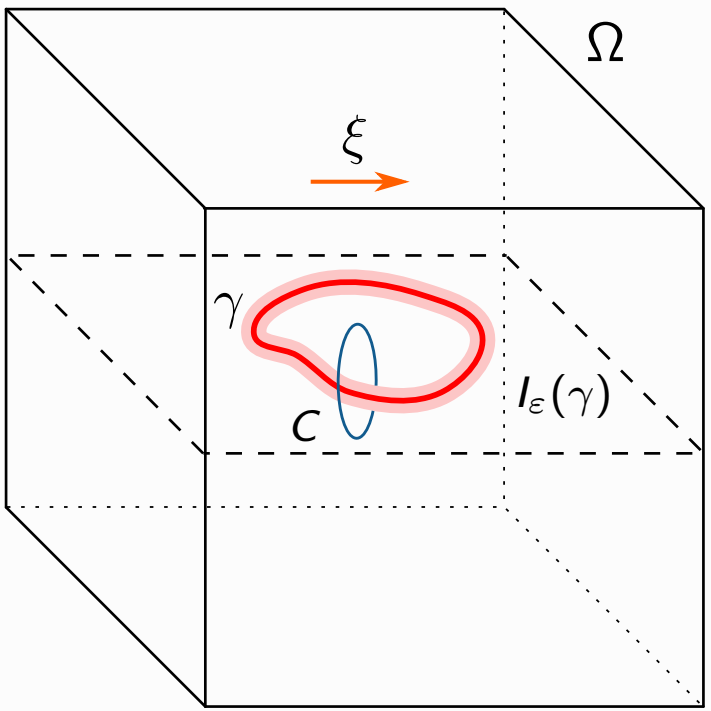
Regularise the problem: Core Radius Approach

Let β generate (γ, ξ). Consider $\varepsilon > 0$ and

$$I_\varepsilon(\gamma) := \{x \in \mathbb{R}^3 : \ \mathrm{dist}(x, \gamma) < \varepsilon\}\,.$$

Then we have

$$|\beta(x)| \sim \frac{1}{\mathrm{dist}(x,\gamma)} \ \text{ in } \ I_\varepsilon(\gamma) \implies \beta \notin L^2(I_\varepsilon(\gamma)).$$



CRA: new ref. conf. $\Omega_\varepsilon(\gamma) := \Omega \setminus I_\varepsilon(\gamma)$.

New Strains: maps $\beta \in L^2(\Omega_\varepsilon(\gamma); \mathbb{M}^{3\times 3})$ s.t.

$$\mathrm{Curl}\, \beta \llcorner \Omega_\varepsilon(\gamma) = 0\,, \qquad \int_C \beta \cdot t \, d\mathcal{H}^1 = \xi\,.$$

Elastic energy associated to β is

$$E_\varepsilon(\beta) := \int_{\Omega_\varepsilon(\gamma)} W(\beta)\, dx\,.$$

Γ-convergence

Let $\mathcal{X}$ be a metric space and $F_n \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$.

Γ-convergence

Let $\mathcal{X}$ be a metric space and $F_n \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$.

Definition (Γ-convergence)

We say that F_n Γ-converges to $F \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$ as $n \to \infty$ if:

Γ-convergence

Let $\mathcal{X}$ be a metric space and $F_n \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$.

Definition (Γ-convergence)

We say that F_n Γ-converges to $F \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$ as $n \to \infty$ if:

- (Γ-liminf inequality) for every $x \in \mathcal{X}$ and every $\{x_n\}$ such that $x_n \to x$,

$$F(x) \leq \liminf_{n\to\infty} F_n(x_n)\,,$$

Γ-convergence

Let $\mathcal{X}$ be a metric space and $F_n \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$.

Definition (Γ-convergence)

We say that F_n Γ-converges to $F \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$ as $n \to \infty$ if:

- (Γ-liminf inequality) for every $x \in \mathcal{X}$ and every $\{x_n\}$ such that $x_n \to x$,
$$F(x) \leq \liminf_{n\to\infty} F_n(x_n)\,,$$
- (Γ-limsup inequality) for every $x \in \mathcal{X}$ there exists a recovery sequence $\{x_n\}$ such that $x_n \to x$ and
$$F(x) = \lim_{n\to\infty} F_n(x_n)\,.$$

Γ-convergence

Let $\mathcal{X}$ be a metric space and $F_n \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$.

Definition (Γ-convergence)

We say that F_n Γ-converges to $F \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$ as $n \to \infty$ if:

- (Γ-liminf inequality) for every $x \in \mathcal{X}$ and every $\{x_n\}$ such that $x_n \to x$,
$$F(x) \leq \liminf_{n\to\infty} F_n(x_n)\,,$$
- (Γ-limsup inequality) for every $x \in \mathcal{X}$ there exists a recovery sequence $\{x_n\}$ such that $x_n \to x$ and
$$F(x) = \lim_{n\to\infty} F_n(x_n)\,.$$

Theorem (Fundamental Theorem of Γ-convergence)

If $F_n \xrightarrow{\Gamma} F$ and $\{x_n\}$ are (almost) minimisers of F_n at each fixed n,

Γ-convergence

Let $\mathcal{X}$ be a metric space and $F_n \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$.

Definition (Γ-convergence)

We say that F_n Γ-converges to $F \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$ as $n \to \infty$ if:

- (Γ-liminf inequality) for every $x \in \mathcal{X}$ and every $\{x_n\}$ such that $x_n \to x$,
$$F(x) \leq \liminf_{n\to\infty} F_n(x_n)\,,$$
- (Γ-limsup inequality) for every $x \in \mathcal{X}$ there exists a recovery sequence $\{x_n\}$ such that $x_n \to x$ and
$$F(x) = \lim_{n\to\infty} F_n(x_n)\,.$$

Theorem (Fundamental Theorem of Γ-convergence)

If $F_n \xrightarrow{\Gamma} F$ and $\{x_n\}$ are (almost) minimisers of F_n at each fixed n,

- *F admits minimum in $\mathcal{X}$ and $\inf_{\mathcal{X}} F_n \to \min_{\mathcal{X}} F$*

Γ-convergence

Let $\mathcal{X}$ be a metric space and $F_n \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$.

Definition (Γ-convergence)

We say that F_n Γ-converges to $F \colon \mathcal{X} \to \mathbb{R} \cup \{\pm\infty\}$ as $n \to \infty$ if:

- (Γ-liminf inequality) for every $x \in \mathcal{X}$ and every $\{x_n\}$ such that $x_n \to x$,
$$F(x) \leq \liminf_{n\to\infty} F_n(x_n)\,,$$
- (Γ-limsup inequality) for every $x \in \mathcal{X}$ there exists a recovery sequence $\{x_n\}$ such that $x_n \to x$ and
$$F(x) = \lim_{n\to\infty} F_n(x_n)\,.$$

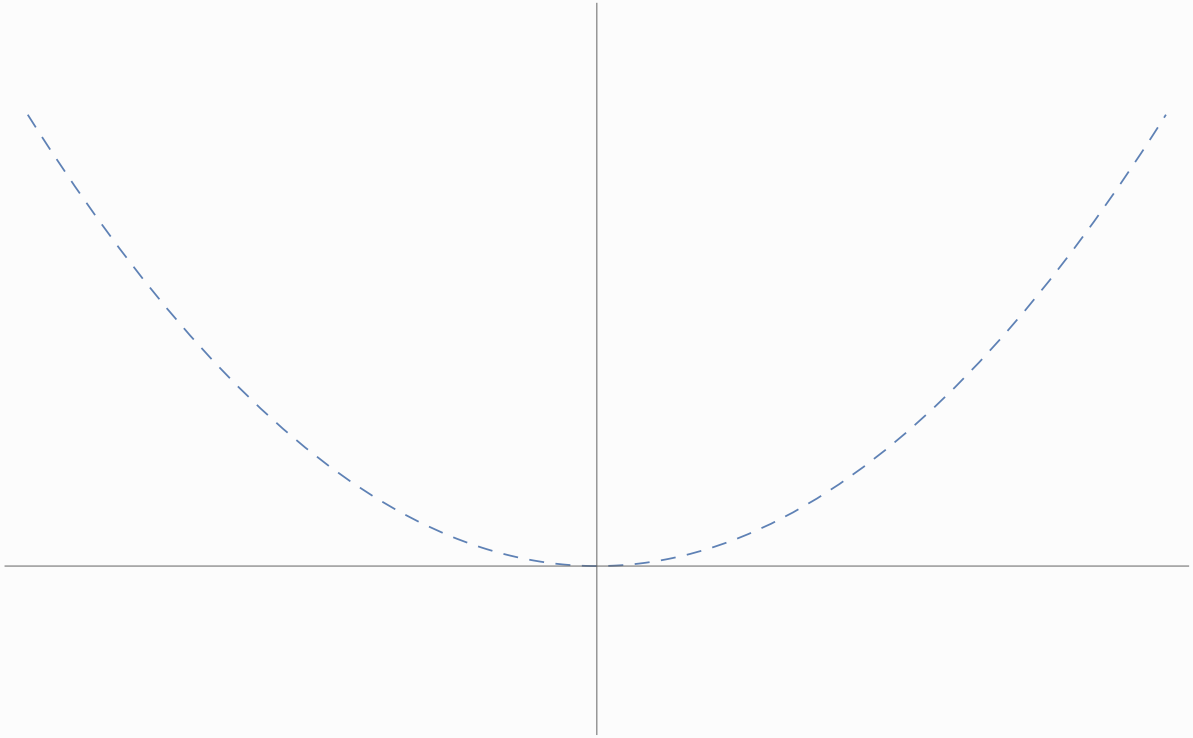
Theorem (Fundamental Theorem of Γ-convergence)

If $F_n \xrightarrow{\Gamma} F$ and $\{x_n\}$ are (almost) minimisers of F_n at each fixed n,

- *F admits minimum in $\mathcal{X}$ and $\inf_{\mathcal{X}} F_n \to \min_{\mathcal{X}} F$*
- *if $x_n \to x$ then $F(x) = \min_{\mathcal{X}} F$.*

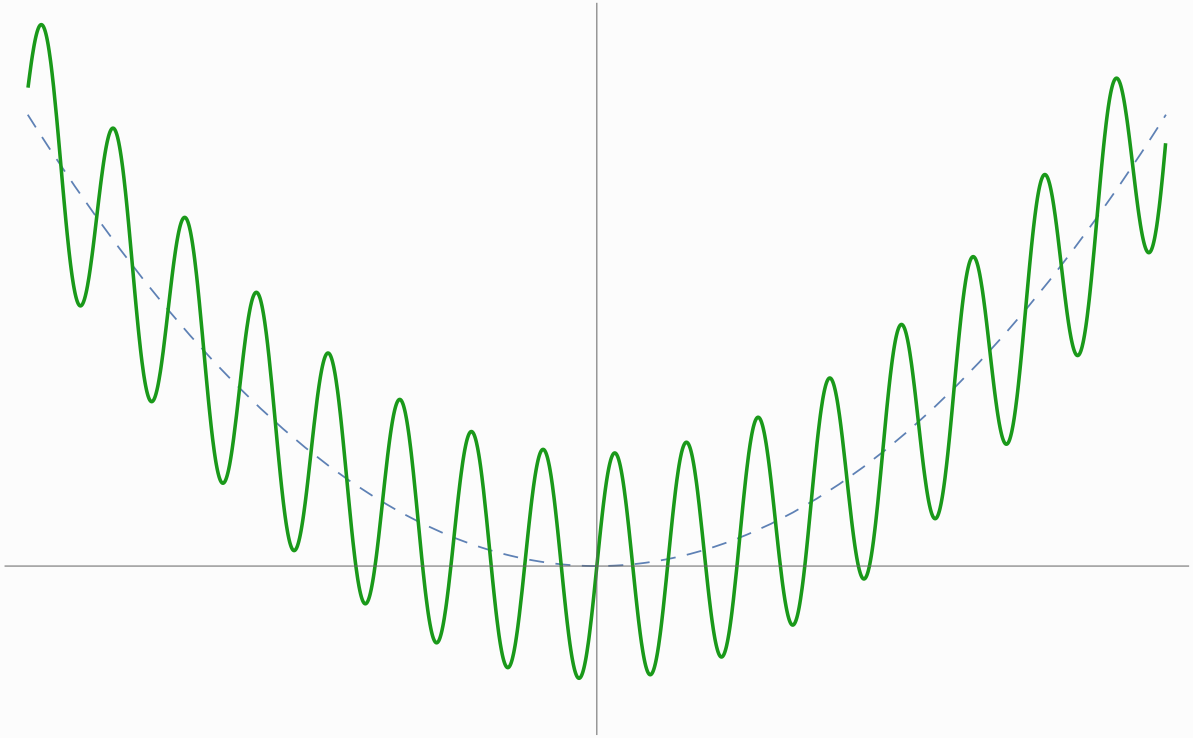
Γ-convergence: basic example

Let $\mathcal{X} = \mathbb{R}$ and define $F_n(x) := x^2 + \cos(nx)$.



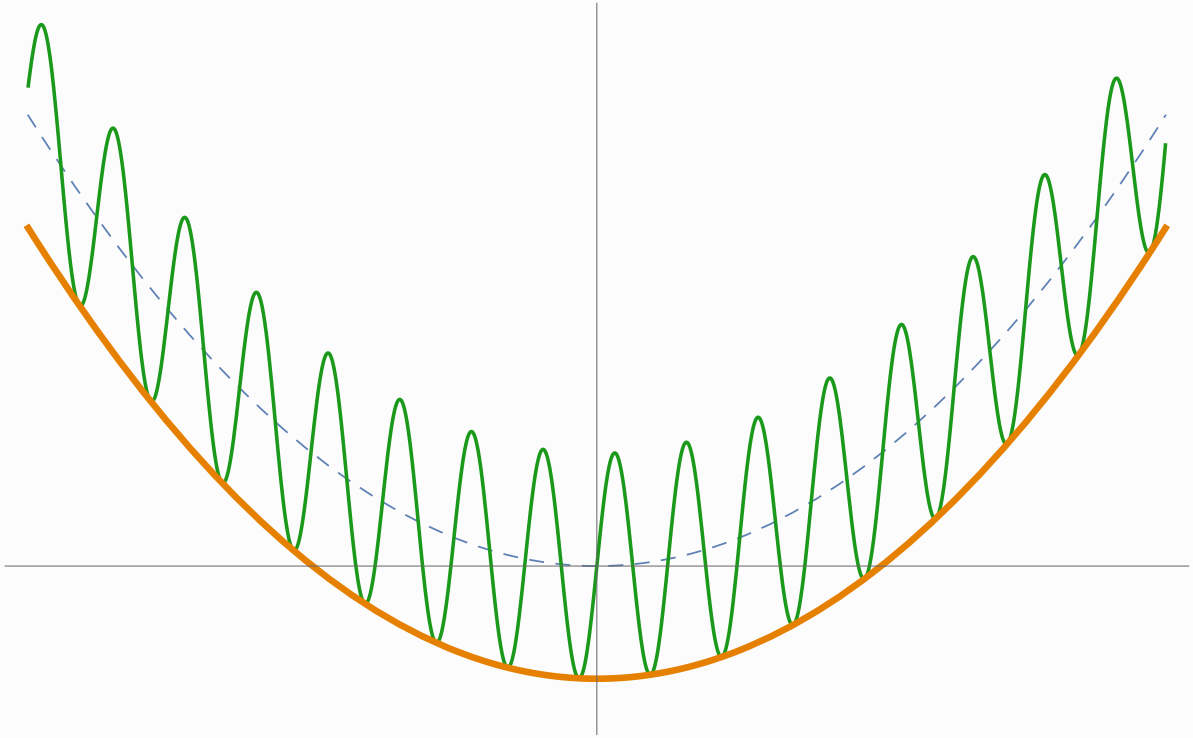
Γ-convergence: basic example

Let $\mathcal{X} = \mathbb{R}$ and define $F_n(x) := x^2 + \cos(nx)$.



Γ-convergence: basic example

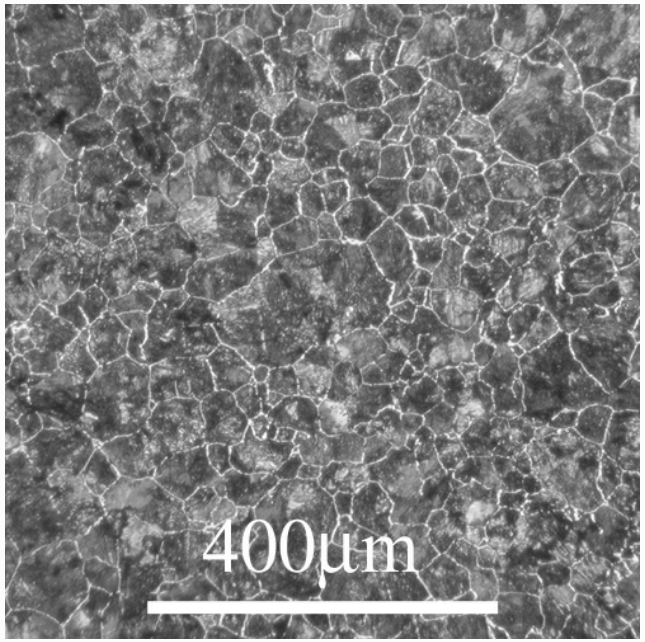
Let $\mathcal{X} = \mathbb{R}$ and define $F_n(x) := x^2 + \cos(nx)$.



We have that $F_n \xrightarrow{\Gamma} F := x^2 - 1$ as $n \to \infty$.

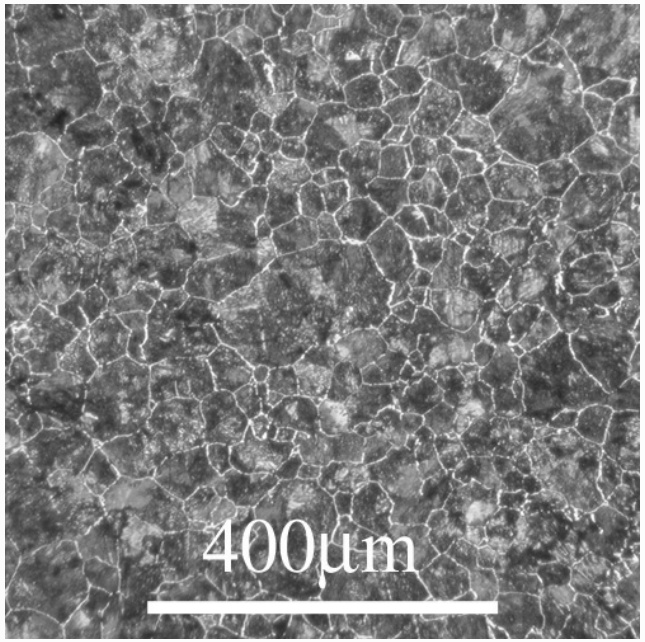
Motivation: polycrystals

Polycrystal: formed by many grains, having the **same** lattice structure, mutually rotated $\implies$ interface misfit at **grain boundaries**.



Motivation: polycrystals

Polycrystal: formed by many grains, having the **same** lattice structure, mutually rotated $\implies$ interface misfit at **grain boundaries**.



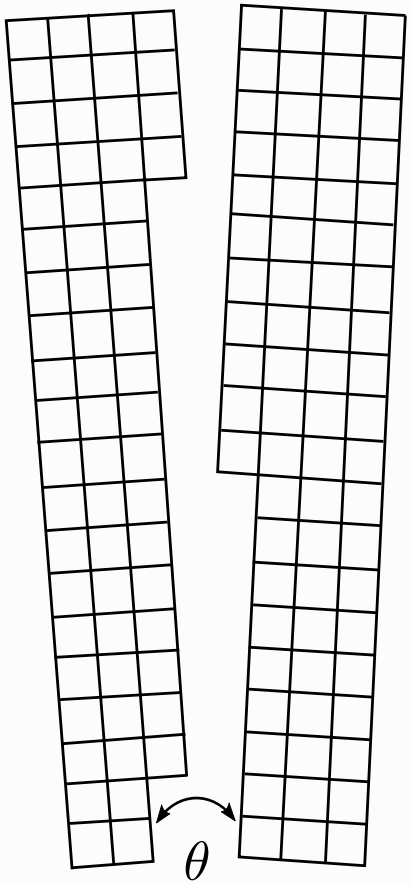
Goal: to obtain polycrystalline structures as minimisers of some energy functional.

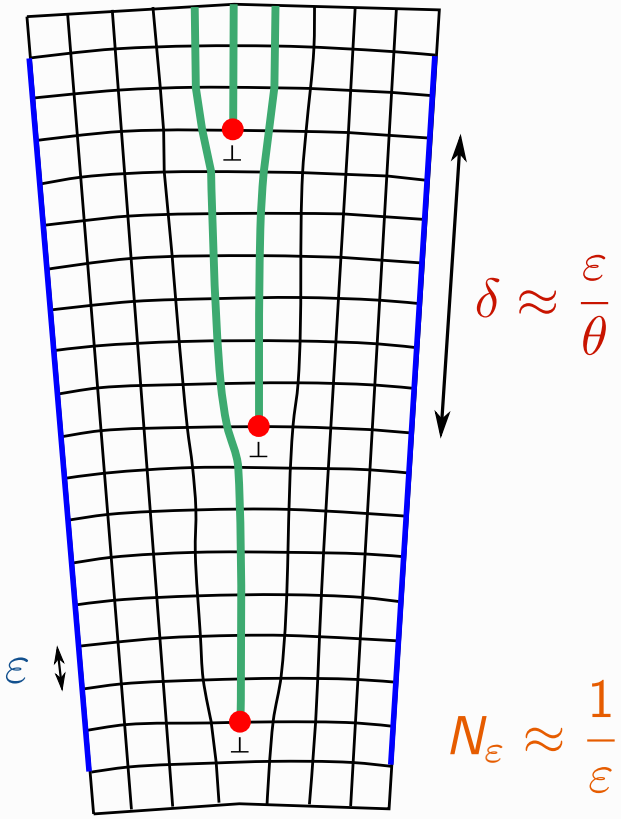
F., Palombaro, Ponsiglione. *Linearised Polycrystals from a 2D System of Edge Dislocations.* Preprint (2017)

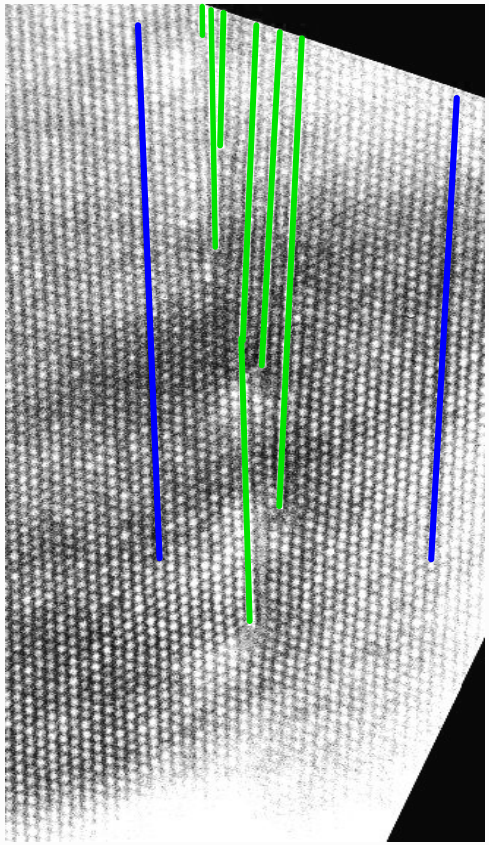
Structure of Tilt Grain Boundaries

Tilt boundary: small angle rotation θ between grains $\implies$ **edge dislocations**.

Boundary structure: periodic array of edge dislocations with spacing $\boldsymbol{\delta = \varepsilon/\theta}$.







Porter, Easterling. CRC Press (2009) - Gottstein. Springer (2013)

Plan of the paper

Setting: consider a 2D system of N_ε edge dislocations, where $\varepsilon > 0$ is the lattice spacing and

$$N_\varepsilon \to +\infty \quad \text{as} \quad \varepsilon \to 0\,.$$

Plan of the paper

Setting: consider a 2D system of N_ε edge dislocations, where $\varepsilon > 0$ is the lattice spacing and

$$N_\varepsilon \to +\infty \quad \text{as} \quad \varepsilon \to 0\,.$$

Plan: let $\mathcal{F}_\varepsilon$ be the energy of such system.

- We compute $\mathcal{F}$, the Γ-limit of $\mathcal{F}_\varepsilon$ as $\varepsilon \to 0$,

Plan of the paper

Setting: consider a 2D system of N_ε edge dislocations, where $\varepsilon > 0$ is the lattice spacing and

$$N_\varepsilon \to +\infty \quad \text{as} \quad \varepsilon \to 0\,.$$

Plan: let $\mathcal{F}_\varepsilon$ be the energy of such system.

- We compute $\mathcal{F}$, the Γ-limit of $\mathcal{F}_\varepsilon$ as $\varepsilon \to 0$,
- we show that under suitable boundary conditions $\mathcal{F}$ is minimised by polycrystals.

Plan of the paper

Setting: consider a 2D system of N_ε edge dislocations, where $\varepsilon > 0$ is the lattice spacing and

$$N_\varepsilon \to +\infty \quad \text{as} \quad \varepsilon \to 0\,.$$

Plan: let $\mathcal{F}_\varepsilon$ be the energy of such system.

- We compute $\mathcal{F}$, the Γ-limit of $\mathcal{F}_\varepsilon$ as $\varepsilon \to 0$,
- we show that under suitable boundary conditions $\mathcal{F}$ is minimised by polycrystals.

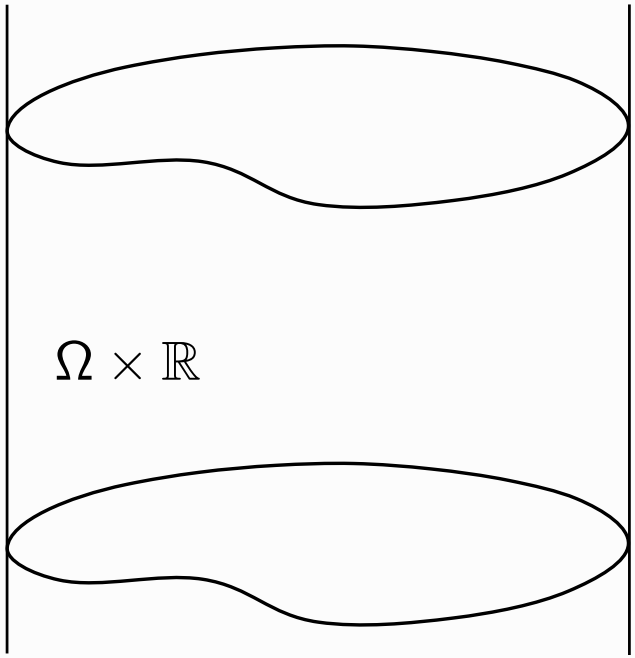
Linearised polycrystals: our energy regime will imply

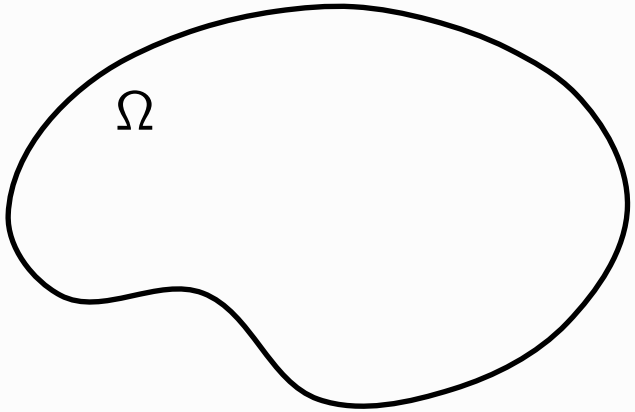
$$N_\varepsilon \ll \frac{1}{\varepsilon}$$

$\implies$ we have less dislocations than tilt grain boundaries. However we still obtain polycrystalline minimisers, but with grains rotated by an infinitesimal angle $\theta \approx 0$.

Setting: linear planar elasticity

Reference configuration: $\Omega \subset \mathbb{R}^2$ open bounded.

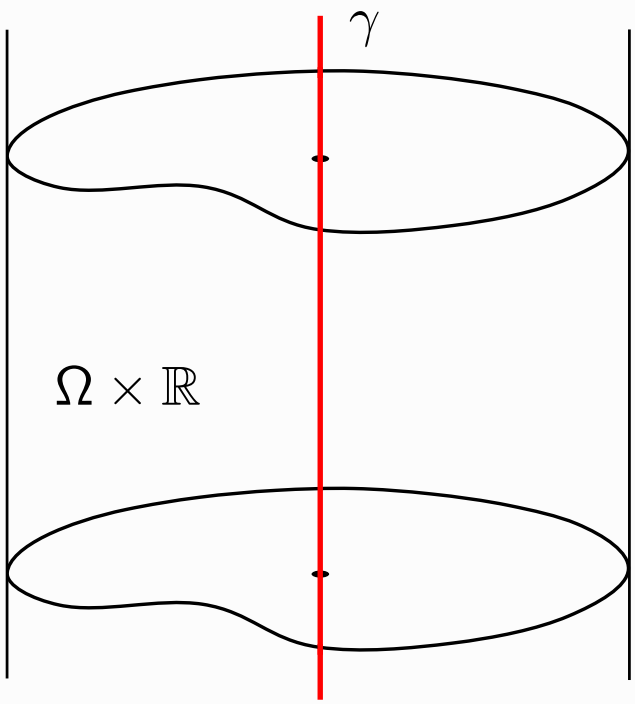


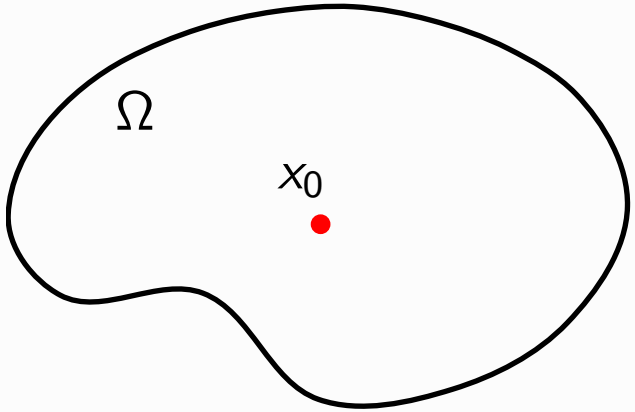


Setting: linear planar elasticity

Reference configuration: $\Omega \subset \mathbb{R}^2$ open bounded.

Dislocation lines: points $x_0 \in \Omega$ separated by 2ε.



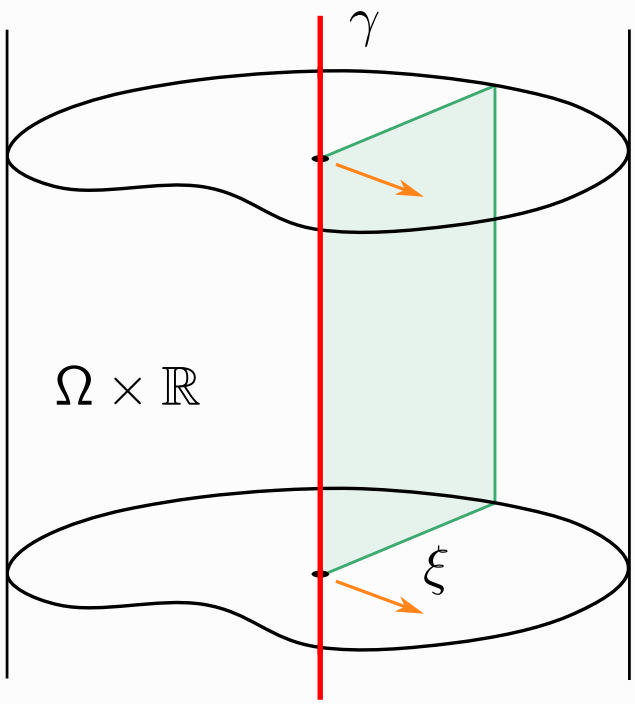


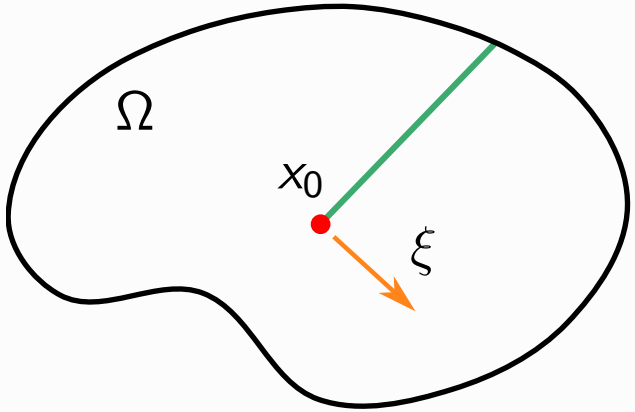
Setting: linear planar elasticity

Reference configuration: $\Omega \subset \mathbb{R}^2$ open bounded.

Dislocation lines: points $x_0 \in \Omega$ separated by 2ε.

Burgers vectors: finite set $\mathcal{S} := \{b_1, \ldots, b_s\} \subset \mathbb{R}^2$.





Setting: linear planar elasticity

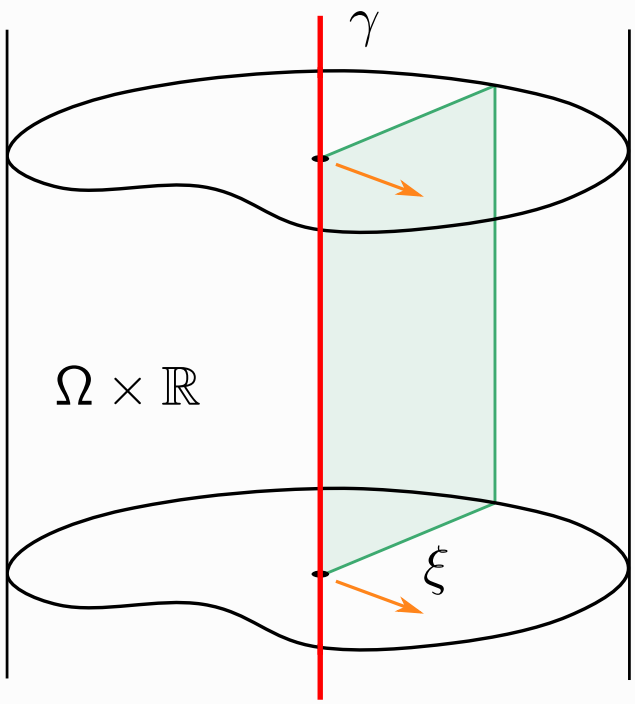
Reference configuration: $\Omega \subset \mathbb{R}^2$ open bounded.

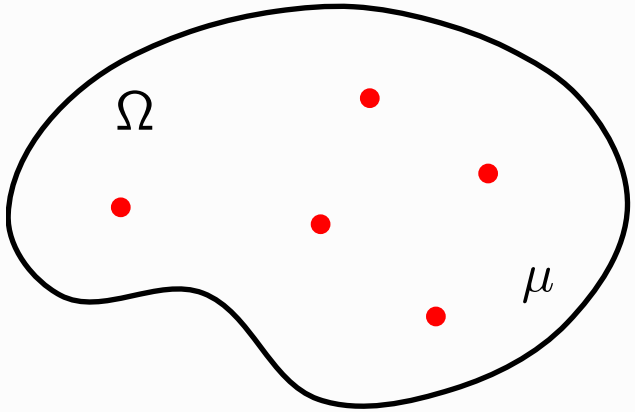
Dislocation lines: points $x_0 \in \Omega$ separated by 2ε.

Burgers vectors: finite set $\mathcal{S} := \{b_1, \dots, b_s\} \subset \mathbb{R}^2$.

Admissible dislocations: finite sums of Dirac masses

$$\mu := \sum_{i=1}^{N} \xi_i \, \delta_{x_i} \,, \quad \xi_i \in \mathcal{S} \,.$$





Setting: linear planar elasticity

Reference configuration: $\Omega \subset \mathbb{R}^2$ open bounded.

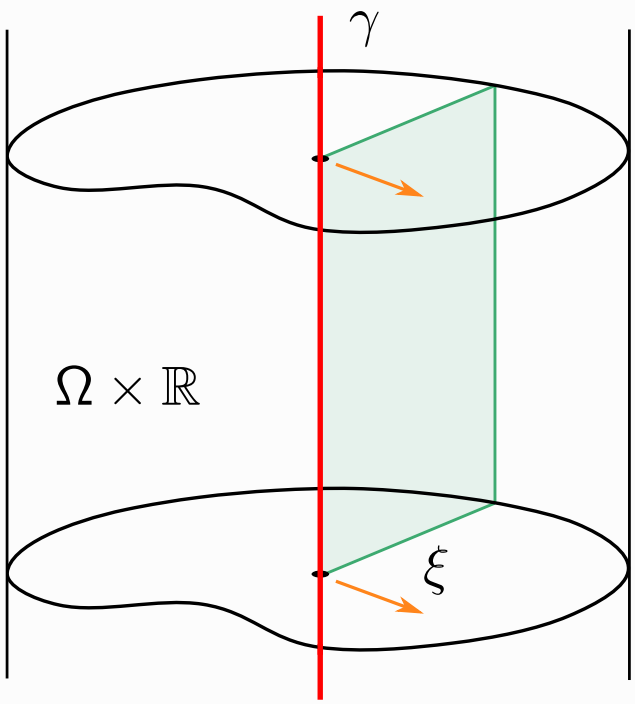
Dislocation lines: points $x_0 \in \Omega$ separated by 2ε.

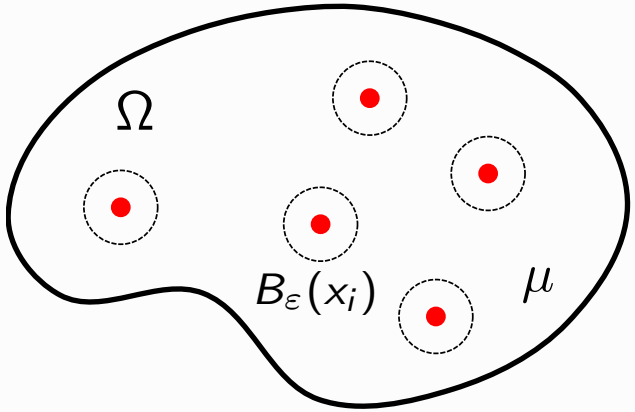
Burgers vectors: finite set $\mathcal{S} := \{b_1, \ldots, b_s\} \subset \mathbb{R}^2$.

Admissible dislocations: finite sums of Dirac masses

$$\mu := \sum_{i=1}^{N} \xi_i \, \delta_{x_i} \,, \quad \xi_i \in \mathcal{S} \,.$$

Core radius approach: $\Omega_\varepsilon(\mu) := \Omega \setminus \cup B_\varepsilon(x_i)$.





Setting: linear planar elasticity

Reference configuration: $\Omega \subset \mathbb{R}^2$ open bounded.

Dislocation lines: points $x_0 \in \Omega$ separated by 2ε.

Burgers vectors: finite set $\mathcal{S} := \{b_1, \dots, b_s\} \subset \mathbb{R}^2$.

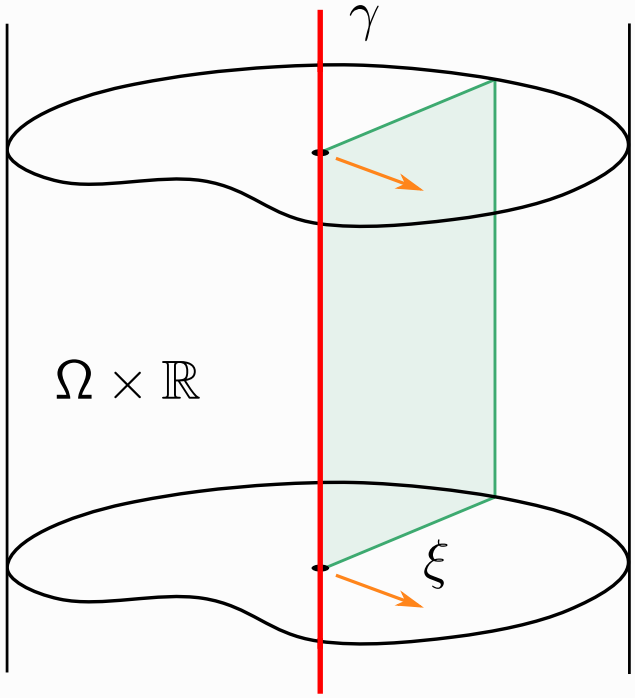
Admissible dislocations: finite sums of Dirac masses

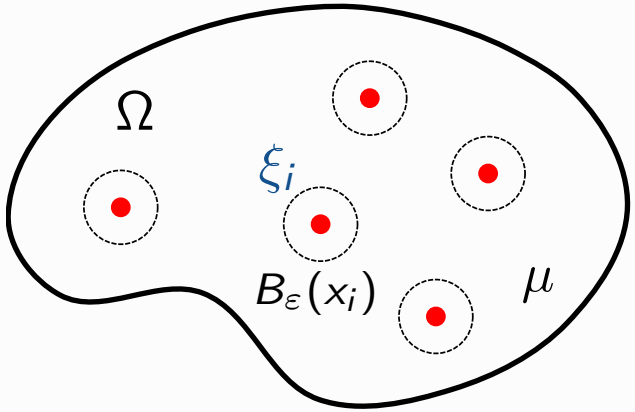
$$\mu := \sum_{i=1}^{N} \xi_i \, \delta_{x_i} \,, \quad \xi_i \in \mathcal{S} \,.$$

Core radius approach: $\Omega_\varepsilon(\mu) := \Omega \setminus \cup B_\varepsilon(x_i)$.

Strains: inducing μ are maps $\beta \colon \Omega_\varepsilon(\mu) \to \mathbb{M}^{2\times 2}$ s.t. $\beta = 0$ in $\cup B_\varepsilon(x_i)$ and

$$\operatorname{Curl} \beta \llcorner \Omega_\varepsilon(\mu) = 0 \,, \qquad \int_{\partial B_\varepsilon(x_i)} \beta \cdot t \, ds = \xi_i \,.$$





Setting: linear planar elasticity

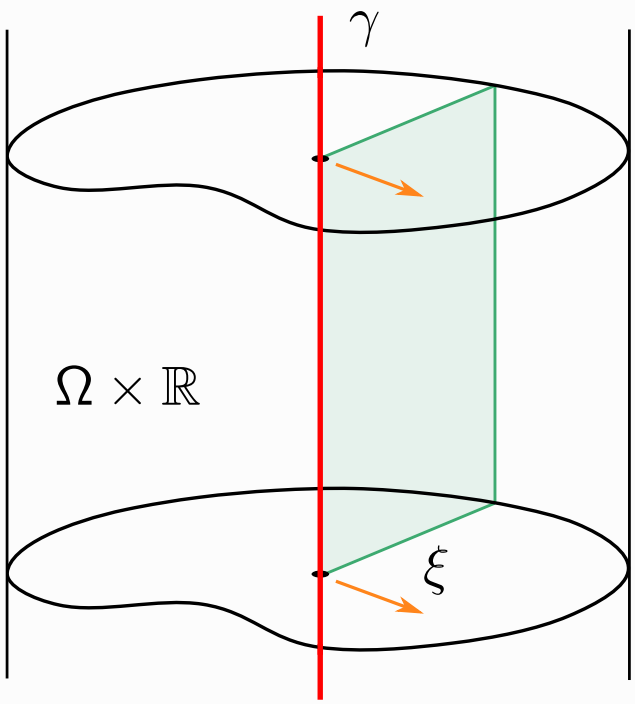
Reference configuration: $\Omega \subset \mathbb{R}^2$ open bounded.

Dislocation lines: points $x_0 \in \Omega$ separated by 2ε.

Burgers vectors: finite set $\mathcal{S} := \{b_1, \dots, b_s\} \subset \mathbb{R}^2$.

Admissible dislocations: finite sums of Dirac masses

$$\mu := \sum_{i=1}^{N} \xi_i \, \delta_{x_i} \,, \quad \xi_i \in \mathcal{S} \,.$$



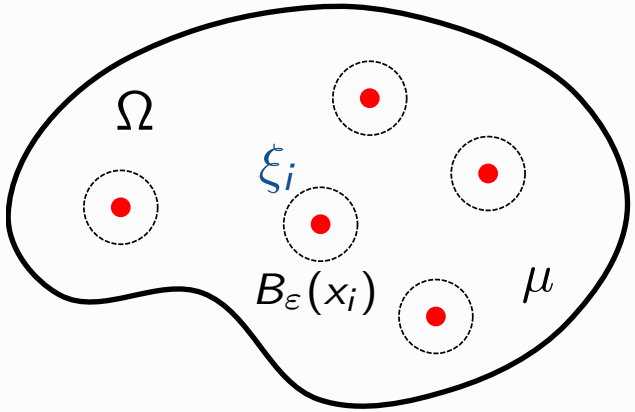
Core radius approach: $\Omega_\varepsilon(\mu) := \Omega \setminus \cup B_\varepsilon(x_i)$.

Strains: inducing μ are maps $\beta \colon \Omega_\varepsilon(\mu) \to \mathbb{M}^{2\times 2}$ s.t. $\beta = 0$ in $\cup B_\varepsilon(x_i)$ and

$$\operatorname{Curl} \beta \llcorner \Omega_\varepsilon(\mu) = 0 \,, \qquad \int_{\partial B_\varepsilon(x_i)} \beta \cdot t \, ds = \xi_i \,.$$

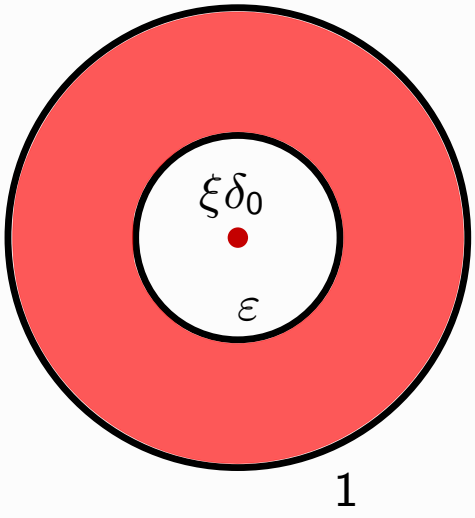
Linear Energy: $\mathbb{C}F : F \sim |F^{\mathrm{sym}}|^2$, then

$$E_\varepsilon(\mu, \beta) := \int_\Omega \mathbb{C}\beta : \beta \, dx = \int_\Omega \mathbb{C}\beta^{\mathrm{sym}} : \beta^{\mathrm{sym}} \, dx \,.$$



Self-energy of a single dislocation core

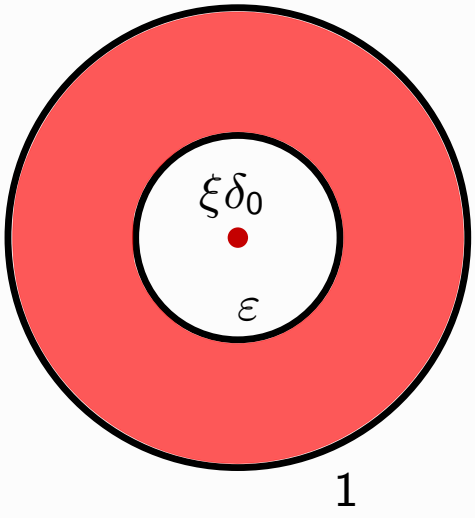
Let β generate $\xi\,\delta_0$, that is "$\mathrm{Curl}\,\beta = \xi\,\delta_0$"



Self-energy of a single dislocation core

Let β generate $\xi\,\delta_0$, that is "$\mathrm{Curl}\,\beta = \xi\,\delta_0$"

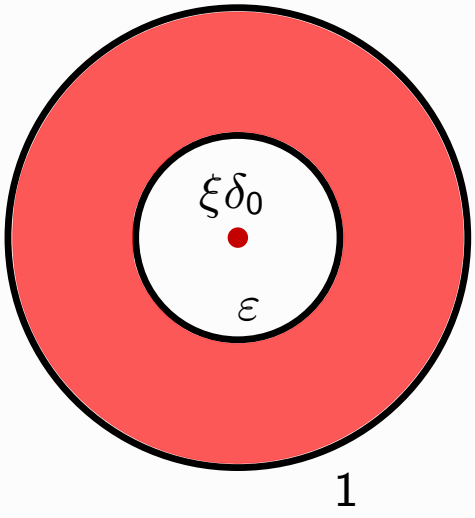
$$\int_{B_1 \setminus B_\varepsilon} |\beta|^2 \, dx$$



Self-energy of a single dislocation core

Let β generate $\xi\,\delta_0$, that is "$\mathrm{Curl}\,\beta = \xi\,\delta_0$"

$$\int_{B_1 \setminus B_\varepsilon} |\beta|^2\, dx \geq \int_\varepsilon^1 \int_{\partial B_\rho} |\beta \cdot t|^2\, ds\, d\rho$$

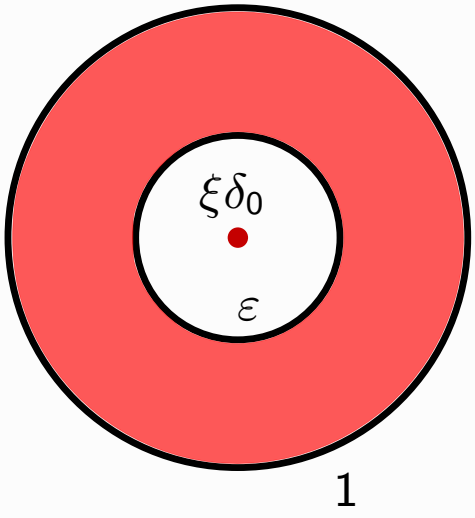


Self-energy of a single dislocation core

Let β generate $\xi\,\delta_0$, that is "$\mathrm{Curl}\,\beta = \xi\,\delta_0$"

$$\int_{B_1\setminus B_\varepsilon} |\beta|^2\,dx \geq \int_\varepsilon^1 \int_{\partial B_\rho} |\beta\cdot t|^2\,ds\,d\rho \geq \text{(Jensen)}$$

$$\geq \frac{1}{2\pi}\int_\varepsilon^1 \frac{1}{\rho}\left|\int_{\partial B_\rho} \beta\cdot t\,ds\right|^2 d\rho$$

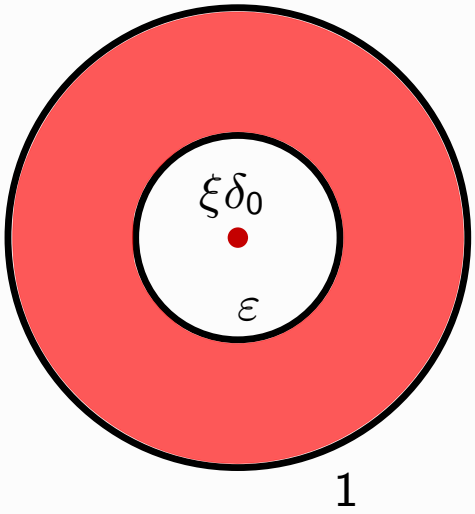


Self-energy of a single dislocation core

Let β generate $\xi\,\delta_0$, that is "$\mathrm{Curl}\,\beta = \xi\,\delta_0$"

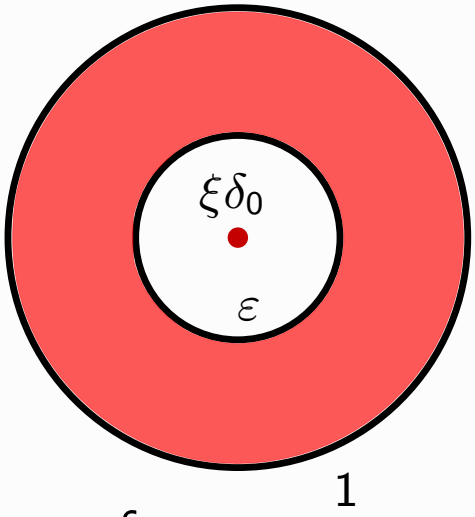
$$\int_{B_1\setminus B_\varepsilon} |\beta|^2\,dx \geq \int_\varepsilon^1 \int_{\partial B_\rho} |\beta\cdot t|^2\,ds\,d\rho \geq \text{(Jensen)}$$

$$\geq \frac{1}{2\pi}\int_\varepsilon^1 \frac{1}{\rho}\left|\int_{\partial B_\rho} \beta\cdot t\,ds\right|^2 d\rho = \frac{|\xi|^2}{2\pi}|\log\varepsilon|\,.$$



Self-energy of a single dislocation core

Let β generate $\xi\,\delta_0$, that is "$\mathrm{Curl}\,\beta = \xi\,\delta_0$"



$$\int_{B_1\setminus B_\varepsilon} |\beta|^2\,dx \geq \int_\varepsilon^1 \int_{\partial B_\rho} |\beta\cdot t|^2\,ds\,d\rho \geq \text{(Jensen)}$$

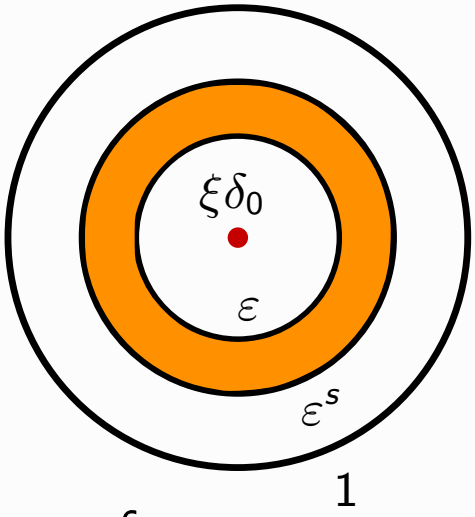
$$\geq \frac{1}{2\pi}\int_\varepsilon^1 \frac{1}{\rho}\left|\int_{\partial B_\rho} \beta\cdot t\,ds\right|^2 d\rho = \frac{|\xi|^2}{2\pi}|\log\varepsilon|\,.$$

The reverse inequality can be obtained by computing the energy of

$$\beta(x) := \frac{1}{2\pi}\,\xi\otimes J\frac{x}{|x|^2}\,,\quad J := \text{clock-wise rotation of } \frac{\pi}{2}\,.$$

Self-energy of a single dislocation core

Let β generate $\xi\,\delta_0$, that is "$\mathrm{Curl}\,\beta = \xi\,\delta_0$"



$$\int_{B_1\setminus B_\varepsilon} |\beta|^2\,dx \geq \int_\varepsilon^1 \int_{\partial B_\rho} |\beta\cdot t|^2\,ds\,d\rho \geq \text{(Jensen)}$$

$$\geq \frac{1}{2\pi}\int_\varepsilon^1 \frac{1}{\rho}\left|\int_{\partial B_\rho} \beta\cdot t\,ds\right|^2 d\rho = \frac{|\xi|^2}{2\pi}|\log\varepsilon|\,.$$

The reverse inequality can be obtained by computing the energy of

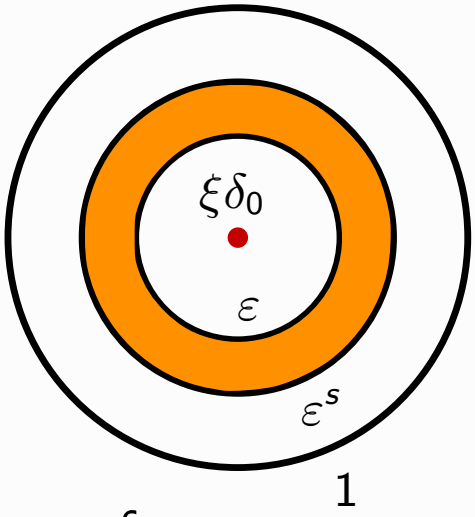
$$\beta(x) := \frac{1}{2\pi}\,\xi\otimes J\frac{x}{|x|^2}\,,\qquad J := \text{clock-wise rotation of } \frac{\pi}{2}\,.$$

Remark: let $s\in(0,1)$, then

$$\int_{B_{\varepsilon^s}\setminus B_\varepsilon} |\beta|^2\,dx \geq (1-s)\frac{|\xi|^2}{2\pi}|\log\varepsilon|\,.$$

Self-energy of a single dislocation core

Let β generate $\xi\,\delta_0$, that is "$\mathrm{Curl}\,\beta = \xi\,\delta_0$"



$$\int_{B_1\setminus B_\varepsilon} |\beta|^2\,dx \geq \int_\varepsilon^1 \int_{\partial B_\rho} |\beta\cdot t|^2\,ds\,d\rho \geq \text{(Jensen)}$$

$$\geq \frac{1}{2\pi}\int_\varepsilon^1 \frac{1}{\rho}\left|\int_{\partial B_\rho} \beta\cdot t\,ds\right|^2 d\rho = \frac{|\xi|^2}{2\pi}|\log\varepsilon|\,.$$

The reverse inequality can be obtained by computing the energy of

$$\beta(x) := \frac{1}{2\pi}\,\xi\otimes J\frac{x}{|x|^2}\,, \quad J := \text{clock-wise rotation of } \frac{\pi}{2}\,.$$

Remark: let $s\in(0,1)$, then

$$\int_{B_{\varepsilon^s}\setminus B_\varepsilon} |\beta|^2\,dx \geq (1-s)\frac{|\xi|^2}{2\pi}|\log\varepsilon|\,.$$

Self-energy: is of order $|\log\varepsilon|$ and concentrated in a small region around B_ε.

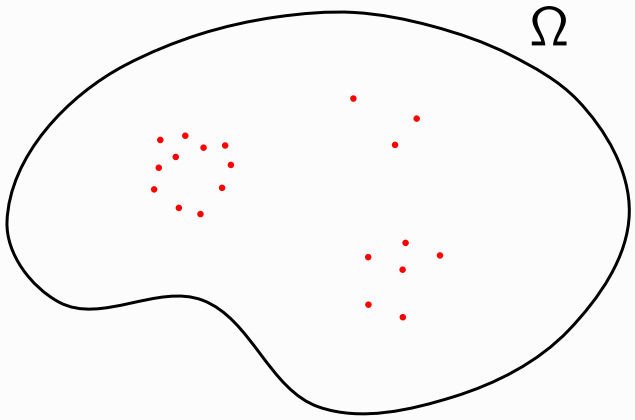
The Hard Core assumption

HC Radius: fixed scale $\rho_\varepsilon \gg \varepsilon$ with $\rho_\varepsilon \to 0$.

The Hard Core assumption

HC Radius: fixed scale $\rho_\varepsilon \gg \varepsilon$ with $\rho_\varepsilon \to 0$.

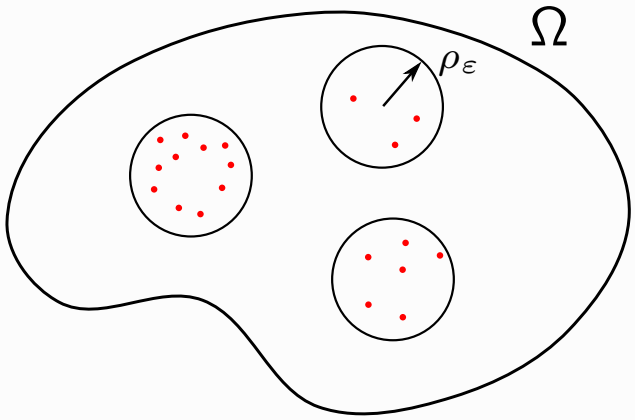
Clusters of dislocations at scale ρ_ε are identified with a single **multiple dislocation**.



The Hard Core assumption

HC Radius: fixed scale $\rho_\varepsilon \gg \varepsilon$ with $\rho_\varepsilon \to 0$.

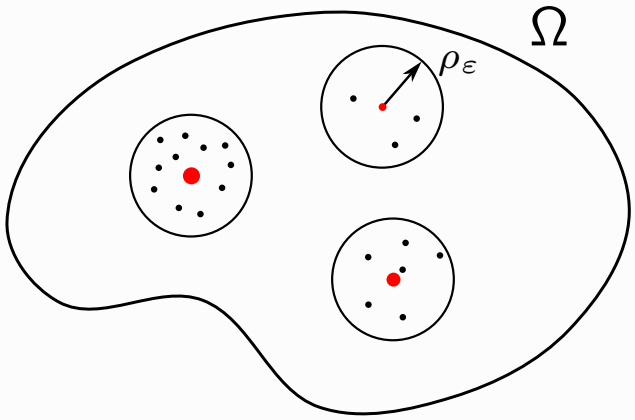
Clusters of dislocations at scale ρ_ε are identified with a single **multiple dislocation**.



The Hard Core assumption

HC Radius: fixed scale $\rho_\varepsilon \gg \varepsilon$ with $\rho_\varepsilon \to 0$.

Clusters of dislocations at scale ρ_ε are identified with a single **multiple dislocation**.



The Hard Core assumption

HC Radius: fixed scale $\rho_\varepsilon \gg \varepsilon$ with $\rho_\varepsilon \to 0$.

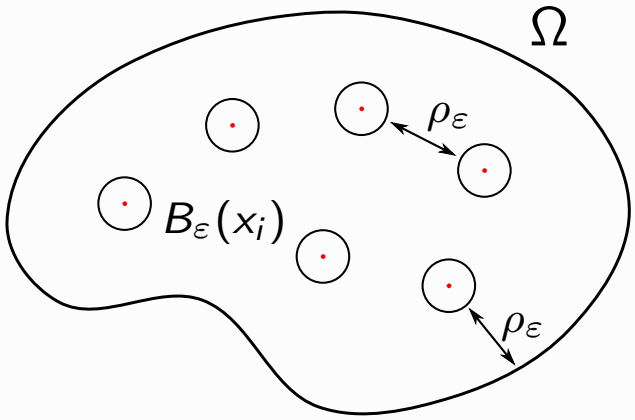
Clusters of dislocations at scale ρ_ε are identified with a single **multiple dislocation**.

Admissible dislocations: finite sums of Dirac masses

$$\mu := \sum_{i=1}^{N} \xi_i \, \delta_{x_i} \,, \quad \xi_i \in \mathbb{S} \,,$$

with $\mathbb{S} := \mathrm{Span}_{\mathbb{Z}} \, \mathcal{S}$ set of multiple Burgers vectors, and

$$|x_i - x_j| > 2\rho_\varepsilon \,, \quad \mathrm{dist}(x_k, \partial\Omega) > \rho_\varepsilon \,.$$



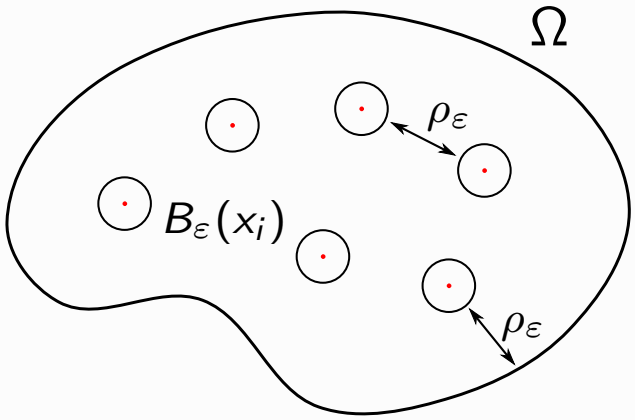
The Hard Core assumption

HC Radius: fixed scale $\rho_\varepsilon \gg \varepsilon$ with $\rho_\varepsilon \to 0$.

Clusters of dislocations at scale ρ_ε are identified with a single **multiple dislocation**.

Admissible dislocations: finite sums of Dirac masses

$$\mu := \sum_{i=1}^{N} \xi_i \, \delta_{x_i} \,, \quad \xi_i \in \mathbb{S} \,,$$



with $\mathbb{S} := \mathrm{Span}_{\mathbb{Z}} \, \mathcal{S}$ set of multiple Burgers vectors, and

$$|x_i - x_j| > 2\rho_\varepsilon \,, \quad \mathrm{dist}(x_k, \partial\Omega) > \rho_\varepsilon \,.$$

Note: dislocations separation is a technical assumption for energy estimates.

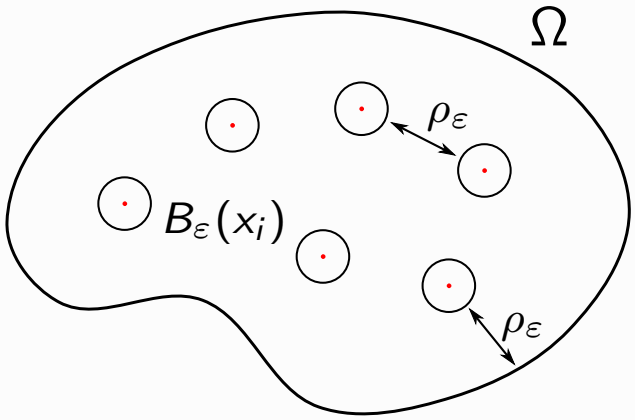
The Hard Core assumption

HC Radius: fixed scale $\rho_\varepsilon \gg \varepsilon$ with $\rho_\varepsilon \to 0$.

Clusters of dislocations at scale ρ_ε are identified with a single **multiple dislocation**.

Admissible dislocations: finite sums of Dirac masses

$$\mu := \sum_{i=1}^{N} \xi_i \, \delta_{x_i} \,, \quad \xi_i \in \mathbb{S} \,,$$



with $\mathbb{S} := \mathrm{Span}_{\mathbb{Z}} \, \mathcal{S}$ set of multiple Burgers vectors, and

$$|x_i - x_j| > 2\rho_\varepsilon \,, \quad \mathrm{dist}(x_k, \partial\Omega) > \rho_\varepsilon \,.$$

Note: dislocations separation is a technical assumption for energy estimates.

Hypothesis on HC Radius: as $\varepsilon \to 0$

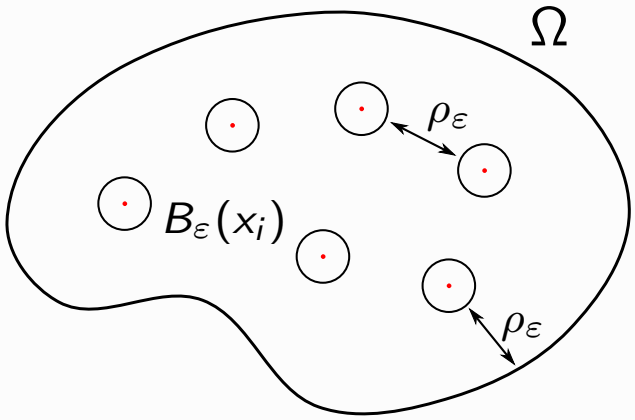
The Hard Core assumption

HC Radius: fixed scale $\rho_\varepsilon \gg \varepsilon$ with $\rho_\varepsilon \to 0$.

Clusters of dislocations at scale ρ_ε are identified with a single **multiple dislocation**.

Admissible dislocations: finite sums of Dirac masses

$$\mu := \sum_{i=1}^{N} \xi_i \, \delta_{x_i} \,, \quad \xi_i \in \mathbb{S} \,,$$



with $\mathbb{S} := \mathrm{Span}_{\mathbb{Z}} \, \mathcal{S}$ set of multiple Burgers vectors, and

$$|x_i - x_j| > 2\rho_\varepsilon \,, \quad \mathrm{dist}(x_k, \partial\Omega) > \rho_\varepsilon \,.$$

Note: dislocations separation is a technical assumption for energy estimates.

Hypothesis on HC Radius: as $\varepsilon \to 0$

- $\rho_\varepsilon / \varepsilon^s \to \infty, \ \forall \, s \in (0,1)$, (HC region contains almost all the self-energy)

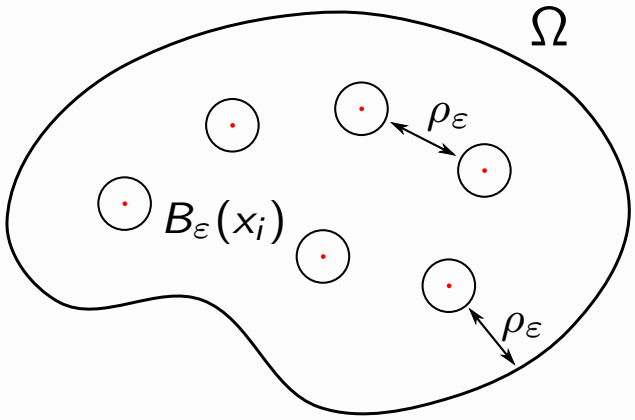
The Hard Core assumption

HC Radius: fixed scale $\rho_\varepsilon \gg \varepsilon$ with $\rho_\varepsilon \to 0$.

Clusters of dislocations at scale ρ_ε are identified with a single **multiple dislocation**.

Admissible dislocations: finite sums of Dirac masses

$$\mu := \sum_{i=1}^{N} \xi_i \, \delta_{x_i} \,, \quad \xi_i \in \mathbb{S} \,,$$



with $\mathbb{S} := \mathrm{Span}_{\mathbb{Z}} \, \mathcal{S}$ set of multiple Burgers vectors, and

$$|x_i - x_j| > 2\rho_\varepsilon \,, \quad \mathrm{dist}(x_k, \partial\Omega) > \rho_\varepsilon \,.$$

Note: dislocations separation is a technical assumption for energy estimates.

Hypothesis on HC Radius: as $\varepsilon \to 0$

- $\rho_\varepsilon / \varepsilon^s \to \infty, \ \forall \, s \in (0,1)$, (HC region contains almost all the self-energy)
- $N_\varepsilon \rho_\varepsilon^2 \to 0$. (Measure of HC region vanishes)

Energy regimes

Energy scaling: each dislocation accounts for $|\log \varepsilon|$ $\implies$ relevant scaling is

$$E_\varepsilon \approx N_\varepsilon |\log \varepsilon| \,.$$

Energy regimes

Energy scaling: each dislocation accounts for $|\log \varepsilon|$ $\implies$ relevant scaling is

$$E_\varepsilon \approx N_\varepsilon |\log \varepsilon| \,.$$

Rescaled energy functionals:

$$\mathcal{F}_\varepsilon(\mu, \beta) := \frac{1}{N_\varepsilon |\log \varepsilon|} \int_{\Omega_\varepsilon(\mu)} \mathbb{C}\beta^{\mathrm{sym}} : \beta^{\mathrm{sym}} \, dx \,.$$

Energy regimes

Energy scaling: each dislocation accounts for $|\log \varepsilon|$ $\implies$ relevant scaling is

$$E_\varepsilon \approx N_\varepsilon |\log \varepsilon| \,.$$

Rescaled energy functionals:

$$\mathcal{F}_\varepsilon(\mu, \beta) := \frac{1}{N_\varepsilon |\log \varepsilon|} \int_{\Omega_\varepsilon(\mu)} \mathbb{C}\beta^{\mathrm{sym}} : \beta^{\mathrm{sym}} \, dx \,.$$

Energy regimes: the behaviour of N_ε determines three different regimes:

- $N_\varepsilon \ll |\log \varepsilon| \rightsquigarrow$ Dilute dislocations

Energy regimes

Energy scaling: each dislocation accounts for $|\log \varepsilon|$ $\implies$ relevant scaling is

$$E_\varepsilon \approx N_\varepsilon |\log \varepsilon| \,.$$

Rescaled energy functionals:

$$\mathcal{F}_\varepsilon(\mu, \beta) := \frac{1}{N_\varepsilon |\log \varepsilon|} \int_{\Omega_\varepsilon(\mu)} \mathbb{C}\beta^{\mathrm{sym}} : \beta^{\mathrm{sym}} \, dx \,.$$

Energy regimes: the behaviour of N_ε determines three different regimes:

- $N_\varepsilon \ll |\log \varepsilon| \rightsquigarrow$ Dilute dislocations
- $N_\varepsilon \approx |\log \varepsilon| \rightsquigarrow$ Critical regime

Garroni, Leoni, Ponsiglione. *Gradient theory for plasticity via homogenization of discrete dislocations.* J. Eur. Math. Soc. (JEMS) (2010)

Energy regimes

Energy scaling: each dislocation accounts for $|\log \varepsilon|$ $\implies$ relevant scaling is

$$E_\varepsilon \approx N_\varepsilon |\log \varepsilon| \,.$$

Rescaled energy functionals:

$$\mathcal{F}_\varepsilon(\mu, \beta) := \frac{1}{N_\varepsilon |\log \varepsilon|} \int_{\Omega_\varepsilon(\mu)} \mathbb{C}\beta^{\mathrm{sym}} : \beta^{\mathrm{sym}} \, dx \,.$$

Energy regimes: the behaviour of N_ε determines three different regimes:

- $N_\varepsilon \ll |\log \varepsilon| \rightsquigarrow$ Dilute dislocations
- $N_\varepsilon \approx |\log \varepsilon| \rightsquigarrow$ Critical regime

  Garroni, Leoni, Ponsiglione. *Gradient theory for plasticity via homogenization of discrete dislocations.* J. Eur. Math. Soc. (JEMS) (2010)

- $N_\varepsilon \gg |\log \varepsilon| \rightsquigarrow$ Super-critical regime

  F., Palombaro, Ponsiglione. *Linearised Polycrystals from a 2D System of Edge Dislocations.* Preprint (2017)

Behaviour of $\mathcal{F}_\varepsilon$ as $\varepsilon \to 0$ (Heuristic)

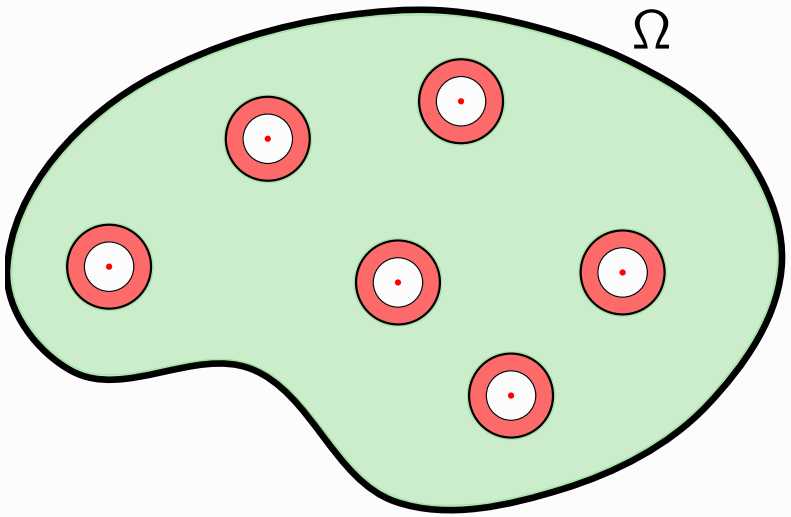
Let (μ, β) with $\mu = \sum_{i=1}^{N_\varepsilon} \xi_i \, \delta_{x_i}$ be such that "$\mathrm{Curl}\, \beta = \mu$".

Behaviour of $\mathcal{F}_\varepsilon$ as $\varepsilon \to 0$ (Heuristic)

Let (μ, β) with $\mu = \sum_{i=1}^{N_\varepsilon} \xi_i \, \delta_{x_i}$ be such that "Curl $\beta = \mu$".

Energy decomposition: let $\mathrm{HC}_\varepsilon(\mu) := \cup_{\mathrm{i}=1}^{\mathrm{N}_\varepsilon} \mathrm{B}_{\rho_\varepsilon}(\mathrm{x_i})$ be the HC region

$$E_\varepsilon(\mu, \beta) = \int_{\Omega \setminus \mathrm{HC}_\varepsilon(\mu)} \mathbb{C}\beta : \beta \, dx + \int_{\mathrm{HC}_\varepsilon(\mu)} \mathbb{C}\beta : \beta \, dx = E_\varepsilon^{\mathrm{interaction}} + E_\varepsilon^{\mathrm{self}} .$$

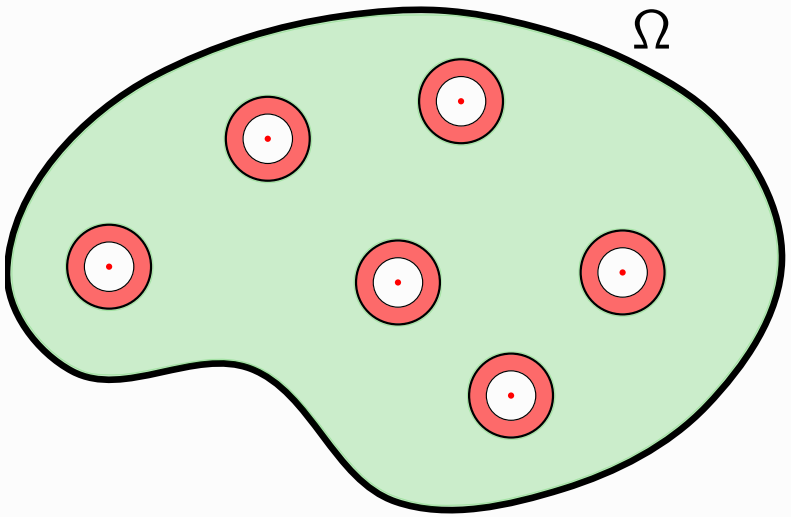


Behaviour of $\mathcal{F}_\varepsilon$ as $\varepsilon \to 0$ (Heuristic)

Let (μ, β) with $\mu = \sum_{i=1}^{N_\varepsilon} \xi_i \, \delta_{x_i}$ be such that "Curl $\beta = \mu$".

Energy decomposition: let $\mathrm{HC}_\varepsilon(\mu) := \cup_{\mathrm{i}=1}^{\mathrm{N}_\varepsilon} \mathrm{B}_{\rho_\varepsilon}(\mathrm{x_i})$ be the HC region

$$E_\varepsilon(\mu, \beta) = \int_{\Omega \setminus \mathrm{HC}_\varepsilon(\mu)} \mathbb{C}\beta : \beta \, dx + \int_{\mathrm{HC}_\varepsilon(\mu)} \mathbb{C}\beta : \beta \, dx = E_\varepsilon^{\mathrm{interaction}} + E_\varepsilon^{\mathrm{self}} .$$



Idea: rescaling by $N_\varepsilon |\log \varepsilon|$, we have $E_\varepsilon^{\mathrm{interaction}} \to E^{\mathrm{elastic}}$ and $E_\varepsilon^{\mathrm{self}} \to E^{\mathrm{plastic}}$.

Γ-convergence result for $N_\varepsilon \gg |\log \varepsilon|$

Theorem (F., Palombaro, Ponsiglione '17)

Compactness: *consider* $(\mu_\varepsilon, \beta_\varepsilon)$ *s.t.* "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" *and* $\mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C \implies$

Γ-convergence result for $N_\varepsilon \gg |\log \varepsilon|$

Theorem (F., Palombaro, Ponsiglione '17)

Compactness: *consider* $(\mu_\varepsilon, \beta_\varepsilon)$ *s.t.* "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" *and* $\mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C \implies$

- $\dfrac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S$, $\dfrac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A$ *in* $L^2(\Omega; \mathbb{M}^{2\times 2})$,

Γ-convergence result for $N_\varepsilon \gg |\log \varepsilon|$

Theorem (F., Palombaro, Ponsiglione '17)

Compactness: *consider* $(\mu_\varepsilon, \beta_\varepsilon)$ *s.t.* "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" *and* $\mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C \implies$

- $\dfrac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S$, $\dfrac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A$ *in* $L^2(\Omega; \mathbb{M}^{2\times 2})$,
- $\dfrac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu$ *in* $\mathcal{M}(\Omega; \mathbb{R}^2)$,

Γ-convergence result for $N_\varepsilon \gg |\log \varepsilon|$

Theorem (F., Palombaro, Ponsiglione '17)

Compactness: *consider* $(\mu_\varepsilon, \beta_\varepsilon)$ *s.t.* "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" *and* $\mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C \implies$

- $\dfrac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S$, $\dfrac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A$ *in* $L^2(\Omega; \mathbb{M}^{2\times 2})$,
- $\dfrac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu$ *in* $\mathcal{M}(\Omega; \mathbb{R}^2)$,
- $\operatorname{Curl} A = \mu$ *and* $\mu \in H^{-1}(\Omega; \mathbb{R}^2)$ $(\implies A \in BV(\Omega; \mathbb{M}^{2\times 2}_{\mathrm{skew}}))$.

Γ-convergence result for $N_\varepsilon \gg |\log \varepsilon|$

Theorem (F., Palombaro, Ponsiglione '17)

Compactness: *consider* $(\mu_\varepsilon, \beta_\varepsilon)$ *s.t.* "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" *and* $\mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C \implies$

- $\dfrac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S\,, \quad \dfrac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A$ *in* $L^2(\Omega; \mathbb{M}^{2\times 2})$,
- $\dfrac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu$ *in* $\mathcal{M}(\Omega; \mathbb{R}^2)$,
- $\operatorname{Curl} A = \mu$ *and* $\mu \in H^{-1}(\Omega; \mathbb{R}^2)$ $(\implies A \in BV(\Omega; \mathbb{M}^{2\times 2}_{\mathrm{skew}}))$.

Γ-convergence: *the functionals* $\mathcal{F}_\varepsilon$ Γ*-converge to*

$$\mathcal{F}(\mu, S, A) := \int_\Omega \mathbb{C}S : S \, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,, \quad \textit{with} \quad \operatorname{Curl} A = \mu\,.$$

Γ-convergence result for $N_\varepsilon \gg |\log \varepsilon|$

Theorem (F., Palombaro, Ponsiglione '17)

Compactness: *consider* $(\mu_\varepsilon, \beta_\varepsilon)$ *s.t.* "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" *and* $\mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C \implies$

- $\dfrac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S\,, \quad \dfrac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A$ *in* $L^2(\Omega; \mathbb{M}^{2\times 2})$,
- $\dfrac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu$ *in* $\mathcal{M}(\Omega; \mathbb{R}^2)$,
- $\operatorname{Curl} A = \mu$ *and* $\mu \in H^{-1}(\Omega; \mathbb{R}^2)$ $(\implies A \in BV(\Omega; \mathbb{M}^{2\times 2}_{\mathrm{skew}}))$.

Γ-convergence: *the functionals* $\mathcal{F}_\varepsilon$ Γ*-converge to*

$$\mathcal{F}(\mu, S, A) := \int_\Omega \mathbb{C}S : S \, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,, \quad \textit{with} \quad \operatorname{Curl} A = \mu\,.$$

Remark:

- S and A live on two different scales with $S \ll A \implies$ terms in $\mathcal{F}$ decoupled.

Γ-convergence result for $N_\varepsilon \gg |\log \varepsilon|$

Theorem (F., Palombaro, Ponsiglione '17)

Compactness: *consider* $(\mu_\varepsilon, \beta_\varepsilon)$ *s.t.* "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" *and* $\mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C \implies$

- $\dfrac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S\,, \quad \dfrac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A$ *in* $L^2(\Omega; \mathbb{M}^{2\times 2})$,
- $\dfrac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu$ *in* $\mathcal{M}(\Omega; \mathbb{R}^2)$,
- $\operatorname{Curl} A = \mu$ *and* $\mu \in H^{-1}(\Omega; \mathbb{R}^2)$ $(\implies A \in BV(\Omega; \mathbb{M}^{2\times 2}_{\mathrm{skew}}))$.

Γ-convergence: *the functionals* $\mathcal{F}_\varepsilon$ Γ*-converge to*

$$\mathcal{F}(\mu, S, A) := \int_\Omega \mathbb{C}S : S \, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,, \quad \textit{with} \quad \operatorname{Curl} A = \mu\,.$$

Remark:

- S and A live on two different scales with $S \ll A \implies$ terms in $\mathcal{F}$ decoupled.
- In the critical regime $N_\varepsilon \approx |\log \varepsilon|$ we have $S \approx A$ and $\operatorname{Curl}(S + A) = \mu$.

The relaxation formula for φ

Self-energy for a single dislocation core $\xi\delta_0$ is

$$\psi(\xi) := \lim_{\varepsilon\to 0} \min_{\beta} \left\{ \frac{1}{|\log\varepsilon|} \int_{B_1\setminus B_\varepsilon} \mathbb{C}\beta : \beta \, dx : \text{ “Curl}\,\beta = \xi\delta_0\text{”} \right\} .$$

The relaxation formula for φ

Self-energy for a single dislocation core $\xi\delta_0$ is

$$\psi(\xi) := \lim_{\varepsilon\to 0} \min_{\beta} \left\{ \frac{1}{|\log\varepsilon|} \int_{B_1\setminus B_\varepsilon} \mathbb{C}\beta : \beta \, dx : \text{ “Curl}\,\beta = \xi\delta_0\text{”} \right\} .$$

Plastic density: the map $\varphi\colon \mathbb{R}^2 \to [0,\infty)$ defined as the relaxation of ψ

$$\varphi(\xi) := \min \left\{ \sum_{i=1}^{M} \lambda_i \psi(\xi_i) : \xi = \sum_{i=1}^{M} \lambda_i \xi_i,\ M \in \mathbb{N},\ \lambda_i \geq 0,\ \xi_i \in \mathbb{S} \right\} .$$

The relaxation formula for φ

Self-energy for a single dislocation core $\xi\delta_0$ is

$$\psi(\xi) := \lim_{\varepsilon\to 0} \min_{\beta} \left\{ \frac{1}{|\log\varepsilon|} \int_{B_1\setminus B_\varepsilon} \mathbb{C}\beta : \beta \, dx : \text{ “Curl}\,\beta = \xi\delta_0\text{”} \right\} .$$

Plastic density: the map $\varphi\colon \mathbb{R}^2 \to [0,\infty)$ defined as the relaxation of ψ

$$\varphi(\xi) := \min \left\{ \sum_{i=1}^{M} \lambda_i \psi(\xi_i) : \ \xi = \sum_{i=1}^{M} \lambda_i \xi_i, \ M \in \mathbb{N}, \ \lambda_i \geq 0, \ \xi_i \in \mathbb{S} \right\} .$$

Note: since the energy is quadratic, in the Γ-limit we have φ instead of ψ.

The relaxation formula for φ

Self-energy for a single dislocation core $\xi\delta_0$ is

$$\psi(\xi) := \lim_{\varepsilon\to 0} \min_{\beta} \left\{ \frac{1}{|\log\varepsilon|} \int_{B_1\setminus B_\varepsilon} \mathbb{C}\beta : \beta \, dx : \text{ “Curl}\,\beta = \xi\delta_0\text{”} \right\} .$$

Plastic density: the map $\varphi\colon \mathbb{R}^2 \to [0,\infty)$ defined as the relaxation of ψ

$$\varphi(\xi) := \min \left\{ \sum_{i=1}^{M} \lambda_i \psi(\xi_i) : \ \xi = \sum_{i=1}^{M} \lambda_i \xi_i, \ M \in \mathbb{N}, \ \lambda_i \geq 0, \ \xi_i \in \mathbb{S} \right\} .$$

Note: since the energy is quadratic, in the Γ-limit we have φ instead of ψ.

Properties: φ is convex and positively 1-homogeneous. Moreover $\exists\ c > 0$ s.t.

$$c^{-1}\,|\xi| \leq \varphi(\xi) \leq c\,|\xi| \,, \qquad \forall\, \xi \in \mathbb{R}^2 .$$

Ideas for compactness: measures

Let $(\mu_\varepsilon, \beta_\varepsilon)$ with $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i} \delta_{x_{\varepsilon,i}}$ and "Curl $\beta_\varepsilon = \mu_\varepsilon$". Assume that

$$\sup_\varepsilon \mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C\,.$$

Ideas for compactness: measures

Let $(\mu_\varepsilon, \beta_\varepsilon)$ with $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i} \delta_{x_{\varepsilon,i}}$ and "Curl $\beta_\varepsilon = \mu_\varepsilon$". Assume that

$$\sup_\varepsilon \mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C\,.$$

We show that

$$\frac{|\mu_\varepsilon|(\Omega)}{N_\varepsilon} = \frac{1}{N_\varepsilon} \sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}| \leq C \implies \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,. \tag{1}$$

Ideas for compactness: measures

Let $(\mu_\varepsilon, \beta_\varepsilon)$ with $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$ and "Curl $\beta_\varepsilon = \mu_\varepsilon$". Assume that

$$\sup_\varepsilon \mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C\,.$$

We show that

$$\frac{|\mu_\varepsilon|(\Omega)}{N_\varepsilon} = \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}| \leq C \implies \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,. \tag{1}$$

$$C \geq \frac{1}{N_\varepsilon |\log \varepsilon|} \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx$$

Ideas for compactness: measures

Let $(\mu_\varepsilon, \beta_\varepsilon)$ with $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$ and "Curl $\beta_\varepsilon = \mu_\varepsilon$". Assume that

$$\sup_\varepsilon \mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C\,.$$

We show that

$$\frac{|\mu_\varepsilon|(\Omega)}{N_\varepsilon} = \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}| \leq C \implies \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,. \tag{1}$$

$$C \geq \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \frac{1}{|\log\varepsilon|}\int_{B_{\rho_\varepsilon}(x_{\varepsilon,i})} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx$$

Ideas for compactness: measures

Let $(\mu_\varepsilon, \beta_\varepsilon)$ with $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$ and "Curl $\beta_\varepsilon = \mu_\varepsilon$". Assume that

$$\sup_\varepsilon \mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C\,.$$

We show that

$$\frac{|\mu_\varepsilon|(\Omega)}{N_\varepsilon} = \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}| \leq C \implies \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,. \tag{1}$$

$$C \geq \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \frac{1}{|\log\varepsilon|}\int_{B_{\rho_\varepsilon}(x_{\varepsilon,i})} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx$$

$$(\rho_\varepsilon \gg \varepsilon) \gtrsim \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \psi(\xi_{\varepsilon,i})$$

Ideas for compactness: measures

Let $(\mu_\varepsilon, \beta_\varepsilon)$ with $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$ and "Curl $\beta_\varepsilon = \mu_\varepsilon$". Assume that

$$\sup_\varepsilon \mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C\,.$$

We show that

$$\frac{|\mu_\varepsilon|(\Omega)}{N_\varepsilon} = \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}| \leq C \implies \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,. \tag{1}$$

$$C \geq \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \frac{1}{|\log\varepsilon|}\int_{B_{\rho_\varepsilon}(x_{\varepsilon,i})} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx$$

$$(\rho_\varepsilon \gg \varepsilon) \gtrsim \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \psi(\xi_{\varepsilon,i}) \geq \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \varphi(\xi_{\varepsilon,i})$$

Ideas for compactness: measures

Let $(\mu_\varepsilon, \beta_\varepsilon)$ with $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$ and "Curl $\beta_\varepsilon = \mu_\varepsilon$". Assume that

$$\sup_\varepsilon \mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C\,.$$

We show that

$$\frac{|\mu_\varepsilon|(\Omega)}{N_\varepsilon} = \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}| \leq C \implies \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,. \tag{1}$$

$$C \geq \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \frac{1}{|\log\varepsilon|}\int_{B_{\rho_\varepsilon}(x_{\varepsilon,i})} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx$$

$$(\rho_\varepsilon \gg \varepsilon) \gtrsim \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \psi(\xi_{\varepsilon,i}) \geq \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \varphi(\xi_{\varepsilon,i}) = \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}|\,\varphi\left(\frac{\xi_{\varepsilon,i}}{|\xi_{\varepsilon,i}|}\right)$$

Ideas for compactness: measures

Let $(\mu_\varepsilon, \beta_\varepsilon)$ with $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$ and "Curl $\beta_\varepsilon = \mu_\varepsilon$". Assume that

$$\sup_\varepsilon \mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C\,.$$

We show that

$$\frac{|\mu_\varepsilon|(\Omega)}{N_\varepsilon} = \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}| \leq C \implies \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,. \tag{1}$$

$$C \geq \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \frac{1}{|\log\varepsilon|}\int_{B_{\rho_\varepsilon}(x_{\varepsilon,i})} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx$$

$$(\rho_\varepsilon \gg \varepsilon) \gtrsim \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \psi(\xi_{\varepsilon,i}) \geq \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \varphi(\xi_{\varepsilon,i}) = \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}|\,\varphi\left(\frac{\xi_{\varepsilon,i}}{|\xi_{\varepsilon,i}|}\right)$$

$$\left(c := \min_{|\xi|=1}\varphi(\xi) > 0\right) \geq \frac{c}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}|$$

Ideas for compactness: measures

Let $(\mu_\varepsilon, \beta_\varepsilon)$ with $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$ and "Curl $\beta_\varepsilon = \mu_\varepsilon$". Assume that

$$\sup_\varepsilon \mathcal{F}_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) \leq C\,.$$

We show that

$$\frac{|\mu_\varepsilon|(\Omega)}{N_\varepsilon} = \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}| \leq C \implies \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,. \tag{1}$$

$$C \geq \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \frac{1}{|\log\varepsilon|}\int_{B_{\rho_\varepsilon}(x_{\varepsilon,i})} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx$$

$$(\rho_\varepsilon \gg \varepsilon) \gtrsim \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \psi(\xi_{\varepsilon,i}) \geq \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \varphi(\xi_{\varepsilon,i}) = \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}|\,\varphi\left(\frac{\xi_{\varepsilon,i}}{|\xi_{\varepsilon,i}|}\right)$$

$$\left(c := \min_{|\xi|=1}\varphi(\xi) > 0\right) \geq \frac{c}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} |\xi_{\varepsilon,i}| = \frac{c}{N_\varepsilon}|\mu_\varepsilon|(\Omega) \implies (1)$$

Ideas for compactness: strains

Symmetric Part: recalling that $\mathbb{C}F : F \geq C|F^{\mathrm{sym}}|^2$, we have

Ideas for compactness: strains

Symmetric Part: recalling that $\mathbb{C}F : F \geq C|F^{\mathrm{sym}}|^2$, we have

$$N_\varepsilon |\log \varepsilon| \geq C \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx \geq C \int_\Omega |\beta_\varepsilon^{\mathrm{sym}}|^2 \, dx$$

Ideas for compactness: strains

Symmetric Part: recalling that $\mathbb{C}F : F \geq C|F^{\mathrm{sym}}|^2$, we have

$$N_\varepsilon |\log \varepsilon| \geq C \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx \geq C \int_\Omega |\beta_\varepsilon^{\mathrm{sym}}|^2 \, dx \implies \frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S \, .$$

Ideas for compactness: strains

Symmetric Part: recalling that $\mathbb{C}F : F \geq C|F^{\mathrm{sym}}|^2$, we have

$$N_\varepsilon |\log \varepsilon| \geq C \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx \geq C \int_\Omega |\beta_\varepsilon^{\mathrm{sym}}|^2 \, dx \implies \frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S \, .$$

Skew Part: we use a Generalised Korn inequality: there exists $C > 0$ s.t. for every $\beta \in L^1(\Omega; \mathbb{M}^{2\times 2})$ with $\operatorname{Curl} \beta \in \mathcal{M}(\Omega; \mathbb{R}^2)$,

$$\int_\Omega |\beta^{\mathrm{skew}}|^2 \, dx \leq C \left(\int_\Omega |\beta^{\mathrm{sym}}|^2 \, dx + |\operatorname{Curl} \beta|(\Omega)^2 \right) . \tag{2}$$

Garroni, Leoni, Ponsiglione. J. Eur. Math. Soc. (JEMS) (2010)

Ideas for compactness: strains

Symmetric Part: recalling that $\mathbb{C}F : F \geq C|F^{\mathrm{sym}}|^2$, we have

$$N_\varepsilon |\log \varepsilon| \geq C \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx \geq C \int_\Omega |\beta_\varepsilon^{\mathrm{sym}}|^2 \, dx \implies \frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S \, .$$

Skew Part: we use a Generalised Korn inequality: there exists $C > 0$ s.t. for every $\beta \in L^1(\Omega; \mathbb{M}^{2\times 2})$ with $\operatorname{Curl} \beta \in \mathcal{M}(\Omega; \mathbb{R}^2)$,

$$\int_\Omega |\beta^{\mathrm{skew}}|^2 \, dx \leq C \left(\int_\Omega |\beta^{\mathrm{sym}}|^2 \, dx + |\operatorname{Curl} \beta|(\Omega)^2 \right) . \tag{2}$$

Since "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$", by (2) and assumption $N_\varepsilon \gg |\log \varepsilon|$ we get

$$\int_\Omega |\beta_\varepsilon^{\mathrm{skew}}|^2 \, dx \leq C \left(\int_\Omega |\beta_\varepsilon^{\mathrm{sym}}|^2 \, dx + |\mu_\varepsilon|(\Omega)^2 \right)$$

.

Garroni, Leoni, Ponsiglione. J. Eur. Math. Soc. (JEMS) (2010)

Ideas for compactness: strains

Symmetric Part: recalling that $\mathbb{C}F : F \geq C|F^{\mathrm{sym}}|^2$, we have

$$N_\varepsilon |\log \varepsilon| \geq C \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx \geq C \int_\Omega |\beta_\varepsilon^{\mathrm{sym}}|^2 \, dx \implies \frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S \, .$$

Skew Part: we use a Generalised Korn inequality: there exists $C > 0$ s.t. for every $\beta \in L^1(\Omega; \mathbb{M}^{2\times 2})$ with $\operatorname{Curl} \beta \in \mathcal{M}(\Omega; \mathbb{R}^2)$,

$$\int_\Omega |\beta^{\mathrm{skew}}|^2 \, dx \leq C \left(\int_\Omega |\beta^{\mathrm{sym}}|^2 \, dx + |\operatorname{Curl} \beta|(\Omega)^2 \right) . \tag{2}$$

Since "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$", by (2) and assumption $N_\varepsilon \gg |\log \varepsilon|$ we get

$$\begin{aligned} \int_\Omega |\beta_\varepsilon^{\mathrm{skew}}|^2 \, dx &\leq C \left(\int_\Omega |\beta_\varepsilon^{\mathrm{sym}}|^2 \, dx + |\mu_\varepsilon|(\Omega)^2 \right) \\ &\leq C \left(N_\varepsilon |\log \varepsilon| + N_\varepsilon^2 \right) \leq C N_\varepsilon^2 \qquad . \end{aligned}$$

Garroni, Leoni, Ponsiglione. J. Eur. Math. Soc. (JEMS) (2010)

Ideas for compactness: strains

Symmetric Part: recalling that $\mathbb{C}F : F \geq C|F^{\mathrm{sym}}|^2$, we have

$$N_\varepsilon |\log \varepsilon| \geq C \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx \geq C \int_\Omega |\beta_\varepsilon^{\mathrm{sym}}|^2 \, dx \implies \frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S \, .$$

Skew Part: we use a Generalised Korn inequality: there exists $C > 0$ s.t. for every $\beta \in L^1(\Omega; \mathbb{M}^{2\times 2})$ with $\operatorname{Curl} \beta \in \mathcal{M}(\Omega; \mathbb{R}^2)$,

$$\int_\Omega |\beta^{\mathrm{skew}}|^2 \, dx \leq C \left(\int_\Omega |\beta^{\mathrm{sym}}|^2 \, dx + |\operatorname{Curl} \beta|(\Omega)^2 \right) . \tag{2}$$

Since "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$", by (2) and assumption $N_\varepsilon \gg |\log \varepsilon|$ we get

$$\begin{aligned} \int_\Omega |\beta_\varepsilon^{\mathrm{skew}}|^2 \, dx &\leq C \left(\int_\Omega |\beta_\varepsilon^{\mathrm{sym}}|^2 \, dx + |\mu_\varepsilon|(\Omega)^2 \right) \\ &\leq C \left(N_\varepsilon |\log \varepsilon| + N_\varepsilon^2 \right) \leq C N_\varepsilon^2 \implies \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A. \end{aligned}$$

Garroni, Leoni, Ponsiglione. J. Eur. Math. Soc. (JEMS) (2010)

Ideas for Γ-liminf

Assume that $(\mu_\varepsilon, \beta_\varepsilon)$ is such that $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$, "Curl $\beta_\varepsilon = \mu_\varepsilon$" and

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon|\log\varepsilon|}} \rightharpoonup S\,, \quad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \quad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \quad \text{with } \operatorname{Curl} A = \mu\,.$$

Ideas for Γ-liminf

Assume that $(\mu_\varepsilon, \beta_\varepsilon)$ is such that $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$, "Curl $\beta_\varepsilon = \mu_\varepsilon$" and

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon|\log\varepsilon|}} \rightharpoonup S\,, \quad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \quad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \quad \text{with } \operatorname{Curl} A = \mu\,.$$

We have to show

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,.$$

Ideas for Γ-liminf

Assume that $(\mu_\varepsilon, \beta_\varepsilon)$ is such that $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$, "Curl $\beta_\varepsilon = \mu_\varepsilon$" and

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon|\log\varepsilon|}} \rightharpoonup S\,, \quad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \quad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \quad \text{with } \operatorname{Curl} A = \mu\,.$$

We have to show

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,.$$

Idea: split the energy $E_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) = E_\varepsilon^{\mathrm{interaction}} + E_\varepsilon^{self}$ and use lower semicontinuity:

Ideas for Γ-liminf

Assume that $(\mu_\varepsilon, \beta_\varepsilon)$ is such that $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$, "Curl $\beta_\varepsilon = \mu_\varepsilon$" and

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon|\log\varepsilon|}} \rightharpoonup S\,, \quad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \quad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \quad \text{with } \operatorname{Curl} A = \mu\,.$$

We have to show

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,.$$

Idea: split the energy $E_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) = E_\varepsilon^{\mathrm{interaction}} + E_\varepsilon^{self}$ and use lower semicontinuity:

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_{\Omega\setminus HC_\varepsilon} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx \quad (N_\varepsilon\rho_\varepsilon^2 \to 0 \implies |HC_\varepsilon| \to 0)$$

Ideas for Γ-liminf

Assume that $(\mu_\varepsilon, \beta_\varepsilon)$ is such that $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$, "Curl $\beta_\varepsilon = \mu_\varepsilon$" and

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon|\log\varepsilon|}} \rightharpoonup S\,, \quad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \quad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \quad \text{with } \operatorname{Curl} A = \mu\,.$$

We have to show

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,.$$

Idea: split the energy $E_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) = E_\varepsilon^{\mathrm{interaction}} + E_\varepsilon^{self}$ and use lower semicontinuity:

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_{\Omega\setminus HC_\varepsilon} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx \quad (N_\varepsilon\rho_\varepsilon^2 \to 0 \implies |HC_\varepsilon| \to 0)$$

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_{HC_\varepsilon} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx$$

Ideas for Γ-liminf

Assume that $(\mu_\varepsilon, \beta_\varepsilon)$ is such that $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$, "Curl $\beta_\varepsilon = \mu_\varepsilon$" and

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon|\log\varepsilon|}} \rightharpoonup S\,, \quad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \quad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \quad \text{with } \operatorname{Curl} A = \mu\,.$$

We have to show

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,.$$

Idea: split the energy $E_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) = E_\varepsilon^{\mathrm{interaction}} + E_\varepsilon^{self}$ and use lower semicontinuity:

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_{\Omega\setminus HC_\varepsilon} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx \quad (N_\varepsilon\rho_\varepsilon^2 \to 0 \implies |HC_\varepsilon| \to 0)$$

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_{HC_\varepsilon} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx = \liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \frac{1}{|\log\varepsilon|}\int_{B_{\rho_\varepsilon}(x_{\varepsilon,i})} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx$$

Ideas for Γ-liminf

Assume that $(\mu_\varepsilon, \beta_\varepsilon)$ is such that $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$, "Curl $\beta_\varepsilon = \mu_\varepsilon$" and

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon|\log\varepsilon|}} \rightharpoonup S\,, \quad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \quad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \quad \text{with } \operatorname{Curl} A = \mu\,.$$

We have to show

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,.$$

Idea: split the energy $E_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) = E_\varepsilon^{\mathrm{interaction}} + E_\varepsilon^{self}$ and use lower semicontinuity:

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_{\Omega\setminus HC_\varepsilon} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx \quad (N_\varepsilon\rho_\varepsilon^2 \to 0 \implies |HC_\varepsilon| \to 0)$$

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_{HC_\varepsilon} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx = \liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \frac{1}{|\log\varepsilon|}\int_{B_{\rho_\varepsilon}(x_{\varepsilon,i})} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx$$

$$\gtrsim \liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \varphi(\xi_{\varepsilon,i})$$

Ideas for Γ-liminf

Assume that $(\mu_\varepsilon, \beta_\varepsilon)$ is such that $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$, "Curl $\beta_\varepsilon = \mu_\varepsilon$" and

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon|\log\varepsilon|}} \rightharpoonup S\,, \quad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \quad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \quad \text{with } \operatorname{Curl} A = \mu\,.$$

We have to show

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,.$$

Idea: split the energy $E_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) = E_\varepsilon^{\mathrm{interaction}} + E_\varepsilon^{self}$ and use lower semicontinuity:

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_{\Omega\setminus HC_\varepsilon} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx \quad (N_\varepsilon\rho_\varepsilon^2 \to 0 \implies |HC_\varepsilon| \to 0)$$

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_{HC_\varepsilon} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx = \liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \frac{1}{|\log\varepsilon|}\int_{B_{\rho_\varepsilon}(x_{\varepsilon,i})} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx$$

$$\gtrsim \liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \varphi(\xi_{\varepsilon,i}) = \liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon}\int_\Omega \varphi\left(\frac{d\mu_\varepsilon}{d|\mu_\varepsilon|}\right) d|\mu_\varepsilon|$$

Ideas for Γ-liminf

Assume that $(\mu_\varepsilon, \beta_\varepsilon)$ is such that $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi_{\varepsilon,i}\delta_{x_{\varepsilon,i}}$, "$\operatorname{Curl}\beta_\varepsilon = \mu_\varepsilon$" and

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon|\log\varepsilon|}} \rightharpoonup S\,, \quad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \quad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \quad \text{with } \operatorname{Curl} A = \mu\,.$$

We have to show

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,.$$

Idea: split the energy $E_\varepsilon(\mu_\varepsilon, \beta_\varepsilon) = E_\varepsilon^{\mathrm{interaction}} + E_\varepsilon^{self}$ and use lower semicontinuity:

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_{\Omega\setminus HC_\varepsilon} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx \geq \int_\Omega \mathbb{C}S : S\, dx \quad (N_\varepsilon\rho_\varepsilon^2 \to 0 \implies |HC_\varepsilon| \to 0)$$

$$\liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon|\log\varepsilon|}\int_{HC_\varepsilon} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx = \liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \frac{1}{|\log\varepsilon|}\int_{B_{\rho_\varepsilon}(x_{\varepsilon,i})} \mathbb{C}\beta_\varepsilon : \beta_\varepsilon\, dx$$

$$\gtrsim \liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon}\sum_{i=1}^{M_\varepsilon} \varphi(\xi_{\varepsilon,i}) = \liminf_{\varepsilon\to 0} \frac{1}{N_\varepsilon}\int_\Omega \varphi\left(\frac{d\mu_\varepsilon}{d|\mu_\varepsilon|}\right) d|\mu_\varepsilon| \geq \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|$$

by Reshetnyak's lower semicontinuity Theorem, since φ is 1-homogeneous.

Ideas for Γ-limsup

Consider (μ, S, A) with S symmetric, A skew and $\operatorname{Curl} A = \mu$.

Ideas for Γ-limsup

Consider (μ, S, A) with S symmetric, A skew and $\operatorname{Curl} A = \mu$. We have to construct a recovery sequence $(\mu_\varepsilon, \beta_\varepsilon)$ with "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" s.t.

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S\,, \qquad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \qquad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \tag{3}$$

$$\lim_{\varepsilon \to 0} \frac{1}{N_\varepsilon |\log \varepsilon|} \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx = \int_\Omega \mathbb{C}S : S \, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,. \tag{4}$$

Ideas for Γ-limsup

Consider (μ, S, A) with S symmetric, A skew and $\operatorname{Curl} A = \mu$. We have to construct a recovery sequence $(\mu_\varepsilon, \beta_\varepsilon)$ with "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" s.t.

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S\,, \quad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \quad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \tag{3}$$

$$\lim_{\varepsilon \to 0} \frac{1}{N_\varepsilon |\log \varepsilon|} \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx = \int_\Omega \mathbb{C}S : S \, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,. \tag{4}$$

For simplicity assume $\mu = \xi \, dx$ with $\varphi(\xi) = \psi(\xi)$. The general is more technical.

Ideas for Γ-limsup

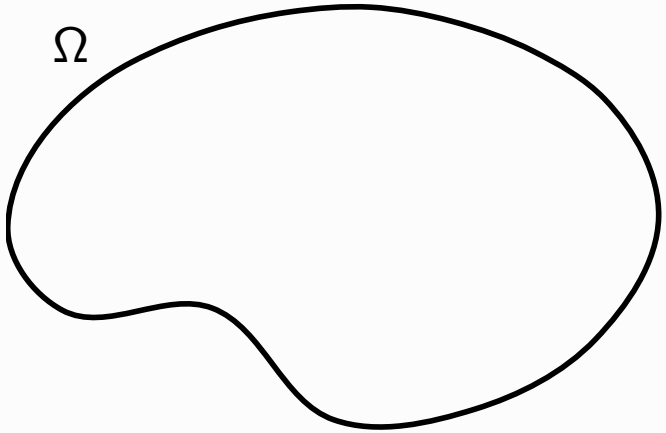
Consider (μ, S, A) with S symmetric, A skew and $\operatorname{Curl} A = \mu$. We have to construct a recovery sequence $(\mu_\varepsilon, \beta_\varepsilon)$ with "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" s.t.

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S\,, \qquad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \qquad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \tag{3}$$

$$\lim_{\varepsilon \to 0} \frac{1}{N_\varepsilon |\log \varepsilon|} \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx = \int_\Omega \mathbb{C}S : S \, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,. \tag{4}$$

For simplicity assume $\mu = \xi \, dx$ with $\varphi(\xi) = \psi(\xi)$. The general is more technical.

Recovery measures:



Ideas for Γ-limsup

Consider (μ, S, A) with S symmetric, A skew and $\operatorname{Curl} A = \mu$. We have to construct a recovery sequence $(\mu_\varepsilon, \beta_\varepsilon)$ with "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" s.t.

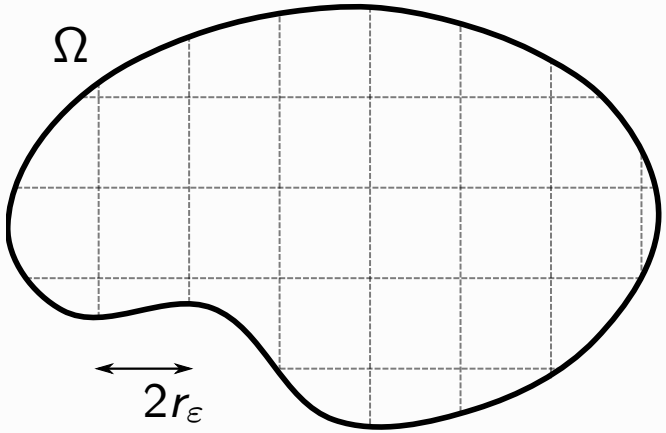
$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S\,, \qquad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \qquad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \tag{3}$$

$$\lim_{\varepsilon \to 0} \frac{1}{N_\varepsilon |\log \varepsilon|} \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx = \int_\Omega \mathbb{C}S : S \, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,. \tag{4}$$

For simplicity assume $\mu = \xi \, dx$ with $\varphi(\xi) = \psi(\xi)$. The general is more technical.

Recovery measures:

- divide Ω in $\approx N_\varepsilon$ squares of side $2r_\varepsilon = C/\sqrt{N_\varepsilon}$



Ideas for Γ-limsup

Consider (μ, S, A) with S symmetric, A skew and $\operatorname{Curl} A = \mu$. We have to construct a recovery sequence $(\mu_\varepsilon, \beta_\varepsilon)$ with "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" s.t.

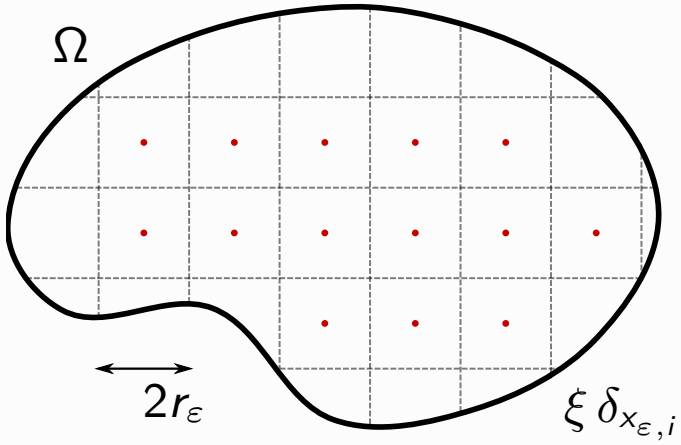
$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S\,, \qquad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \qquad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \tag{3}$$

$$\lim_{\varepsilon \to 0} \frac{1}{N_\varepsilon |\log \varepsilon|} \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx = \int_\Omega \mathbb{C}S : S \, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,. \tag{4}$$

For simplicity assume $\mu = \xi \, dx$ with $\varphi(\xi) = \psi(\xi)$. The general is more technical.

Recovery measures:

- divide Ω in $\approx N_\varepsilon$ squares of side $2r_\varepsilon = C/\sqrt{N_\varepsilon}$
- the recovery sequence is $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi \delta_{\varepsilon,i}$.



Ideas for Γ-limsup

Consider (μ, S, A) with S symmetric, A skew and $\operatorname{Curl} A = \mu$. We have to construct a recovery sequence $(\mu_\varepsilon, \beta_\varepsilon)$ with "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" s.t.

$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S\,, \qquad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \qquad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \tag{3}$$

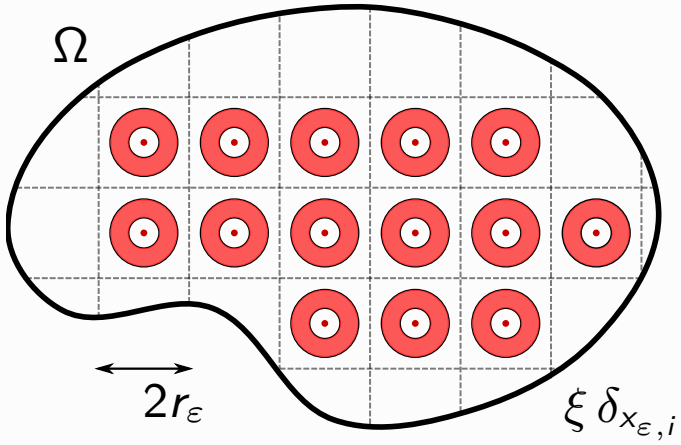
$$\lim_{\varepsilon \to 0} \frac{1}{N_\varepsilon |\log \varepsilon|} \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx = \int_\Omega \mathbb{C}S : S \, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,. \tag{4}$$

For simplicity assume $\mu = \xi \, dx$ with $\varphi(\xi) = \psi(\xi)$. The general is more technical.

Recovery measures:

- divide Ω in $\approx N_\varepsilon$ squares of side $2r_\varepsilon = C/\sqrt{N_\varepsilon}$
- the recovery sequence is $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi \delta_{\varepsilon,i}$.

Recovery strains: let K_ε be the solution to the cell-problem about each $x_{\varepsilon,i}$.



Ideas for Γ-limsup

Consider (μ, S, A) with S symmetric, A skew and $\operatorname{Curl} A = \mu$. We have to construct a recovery sequence $(\mu_\varepsilon, \beta_\varepsilon)$ with "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$" s.t.

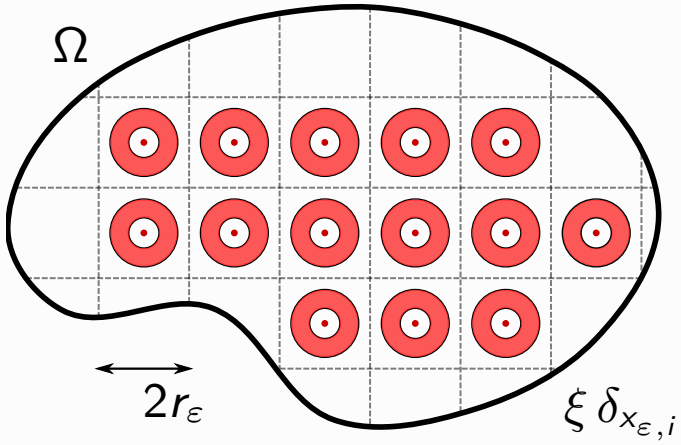
$$\frac{\beta_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup S\,, \qquad \frac{\beta_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup A\,, \qquad \frac{\mu_\varepsilon}{N_\varepsilon} \overset{*}{\rightharpoonup} \mu\,, \tag{3}$$

$$\lim_{\varepsilon\to 0} \frac{1}{N_\varepsilon |\log \varepsilon|} \int_\Omega \mathbb{C}\beta_\varepsilon : \beta_\varepsilon \, dx = \int_\Omega \mathbb{C}S : S \, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu|\,. \tag{4}$$

For simplicity assume $\mu = \xi \, dx$ with $\varphi(\xi) = \psi(\xi)$. The general is more technical.

Recovery measures:

- divide Ω in $\approx N_\varepsilon$ squares of side $2r_\varepsilon = C/\sqrt{N_\varepsilon}$
- the recovery sequence is $\mu_\varepsilon = \sum_{i=1}^{M_\varepsilon} \xi \delta_{\varepsilon,i}$.



Recovery strains: let K_ε be the solution to the cell-problem about each $x_{\varepsilon,i}$. Then $K_\varepsilon/\sqrt{N_\varepsilon|\log\varepsilon|} \rightharpoonup 0$ and

$$\beta_\varepsilon = \sqrt{N_\varepsilon|\log\varepsilon|}S + N_\varepsilon A + K_\varepsilon + O(\sqrt{N_\varepsilon|\log\varepsilon|})$$

satisfies (3), (4) and "$\operatorname{Curl} \beta_\varepsilon = \mu_\varepsilon$".

Adding boundary conditions

Dirichlet type BC: at level $\varepsilon > 0$ fix a boundary condition $g_\varepsilon \colon \Omega \to \mathbb{M}^{2\times 2}$ s.t.

$$\frac{g_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup g_S \,, \qquad \frac{g_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup g_A \,.$$

Adding boundary conditions

Dirichlet type BC: at level $\varepsilon > 0$ fix a boundary condition $g_\varepsilon \colon \Omega \to \mathbb{M}^{2\times 2}$ s.t.

$$\frac{g_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup g_S \,, \qquad \frac{g_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup g_A \,.$$

Dislocations and strains: (μ, β) such that " $\mathrm{Curl}\, \beta = \mu$" and

$$\mu(\Omega) = \int_{\partial\Omega} g_\varepsilon \cdot t \, ds \,, \qquad \beta \cdot t = g_\varepsilon \cdot t \quad \text{on} \quad \partial\Omega \,.$$

Adding boundary conditions

Dirichlet type BC: at level $\varepsilon > 0$ fix a boundary condition $g_\varepsilon \colon \Omega \to \mathbb{M}^{2\times 2}$ s.t.

$$\frac{g_\varepsilon^{\mathrm{sym}}}{\sqrt{N_\varepsilon |\log \varepsilon|}} \rightharpoonup g_S \,, \qquad \frac{g_\varepsilon^{\mathrm{skew}}}{N_\varepsilon} \rightharpoonup g_A \,.$$

Dislocations and strains: (μ, β) such that " $\mathrm{Curl}\, \beta = \mu$" and

$$\mu(\Omega) = \int_{\partial\Omega} g_\varepsilon \cdot t \, ds \,, \qquad \beta \cdot t = g_\varepsilon \cdot t \quad \text{on} \quad \partial\Omega \,.$$

Theorem (F., Palombaro, Ponsiglione '17)

The energy functionals $\mathcal{F}_\varepsilon$ are equi-coercive and they Γ-converge to

$$\mathcal{F}_{\mathrm{BC}}(\mu, S, A) := \int_\Omega \mathbb{C}S : S \, dx + \int_\Omega \varphi\left(\frac{d\mu}{d|\mu|}\right) d|\mu| + \int_{\partial\Omega} \varphi((g_A - A) \cdot t) \, ds \,,$$

with $\mathrm{Curl}\, A = \mu$ and $\mu \in \mathcal{M}(\Omega; \mathbb{R}^2) \cap H^{-1}(\Omega; \mathbb{R}^2)$.

Remark: $\beta^{\mathrm{sym}} \ll \beta^{\mathrm{skew}} \implies$ BC pass to the limit only for A.

Minimising $\mathcal{F}_{\mathrm{BC}}$ with piecewise constant BC

Remark: there are no BC on S $\implies$ we can neglect elastic energy.

Minimising $\mathcal{F}_{\mathrm{BC}}$ with piecewise constant BC

Remark: there are no BC on $S \implies$ we can neglect elastic energy.

Piecewise constant BC: Fix a piecewise constant BC

$$g_A := \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}, \quad a := \sum_{k=1}^{M} m_k \, \chi_{U_k} \, ,$$

with $m_k < m_{k+1}$ and $\{U_k\}_{k=1}^{M}$ Caccioppoli partition of Ω.

Minimising $\mathcal{F}_{\mathrm{BC}}$ with piecewise constant BC

Remark: there are no BC on S $\implies$ we can neglect elastic energy.

Piecewise constant BC: Fix a piecewise constant BC

$$g_A := \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}, \quad a := \sum_{k=1}^{M} m_k \, \chi_{U_k} \,,$$

with $m_k < m_{k+1}$ and $\{U_k\}_{k=1}^{M}$ Caccioppoli partition of Ω.

Problem

Minimise

$$\mathcal{F}_{\mathrm{BC}}(\mathrm{Curl}\, A, 0, A) = \int_\Omega \varphi\left(\frac{d\,\mathrm{Curl}\, A}{d|\,\mathrm{Curl}\, A|}\right) d|\,\mathrm{Curl}\, A| + \int_{\partial\Omega} \varphi((g_A - A)\cdot t)\, ds \,,$$

with $\mathrm{Curl}\, A \in \mathcal{M}(\Omega;\mathbb{R}^2) \cap H^{-1}(\Omega;\mathbb{R}^2)$.

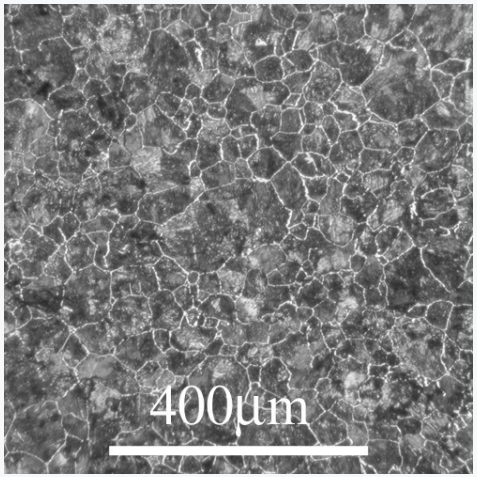
Polycrystals as energy minimisers

Theorem (F., Palombaro, Ponsiglione '17)

Given a piecewise constant boundary condition g_A, there exists a piecewise constant minimiser of $\mathcal{F}_{\mathrm{BC}}(\mathrm{Curl}\, A, 0, A)$

$$A = \sum_{k=1}^{M} A_k \chi_{E_k} \,,$$

with $A_k \in \mathbb{M}^{2\times 2}_{\mathrm{skew}}$ and $\{E_k\}_{k=1}^{M}$ Caccioppoli partition of Ω. We interpret A as a linearised polycrystal.



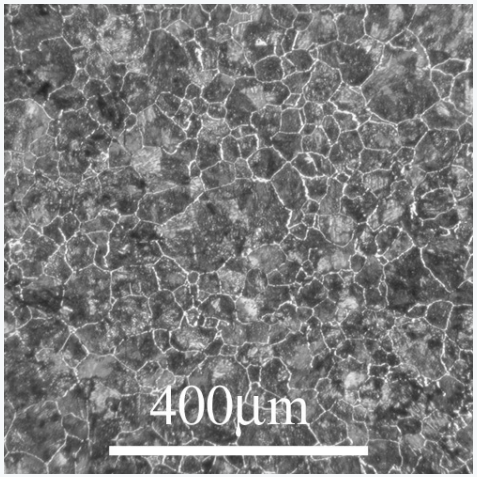
Polycrystals as energy minimisers

Theorem (F., Palombaro, Ponsiglione '17)

Given a piecewise constant boundary condition g_A, there exists a piecewise constant minimiser of $\mathcal{F}_{\mathrm{BC}}(\mathrm{Curl}\, A, 0, A)$

$$A = \sum_{k=1}^{M} A_k \chi_{E_k} \,,$$

with $A_k \in \mathbb{M}^{2\times 2}_{\mathrm{skew}}$ and $\{E_k\}_{k=1}^{M}$ Caccioppoli partition of Ω. We interpret A as a linearised polycrystal.



Proof Strategy: We are minimising an anisotropic total variation functional. By Coarea formula we select the levels with minimal perimeter, definying the Caccioppoli partition.

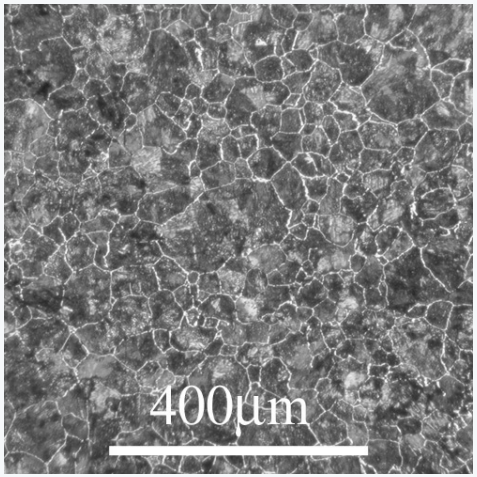
Polycrystals as energy minimisers

Theorem (F., Palombaro, Ponsiglione '17)

Given a piecewise constant boundary condition g_A, there exists a piecewise constant minimiser of $\mathcal{F}_{\mathrm{BC}}(\mathrm{Curl}\, A, 0, A)$

$$A = \sum_{k=1}^{M} A_k \chi_{E_k} \,,$$

with $A_k \in \mathbb{M}^{2\times 2}_{\mathrm{skew}}$ and $\{E_k\}_{k=1}^{M}$ Caccioppoli partition of Ω. We interpret A as a linearised polycrystal.



Proof Strategy: We are minimising an anisotropic total variation functional. By Coarea formula we select the levels with minimal perimeter, definying the Caccioppoli partition.

Open Question: Are all minimisers piecewise constant? Uniqueness?

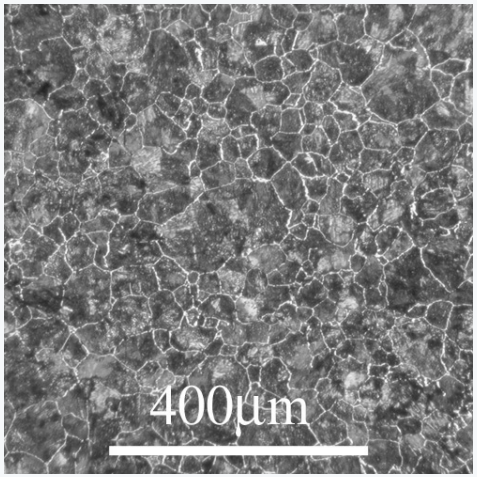
Polycrystals as energy minimisers

Theorem (F., Palombaro, Ponsiglione '17)

Given a piecewise constant boundary condition g_A, there exists a piecewise constant minimiser of $\mathcal{F}_{\mathrm{BC}}(\mathrm{Curl}\, A, 0, A)$

$$A = \sum_{k=1}^{M} A_k \chi_{E_k} \,,$$

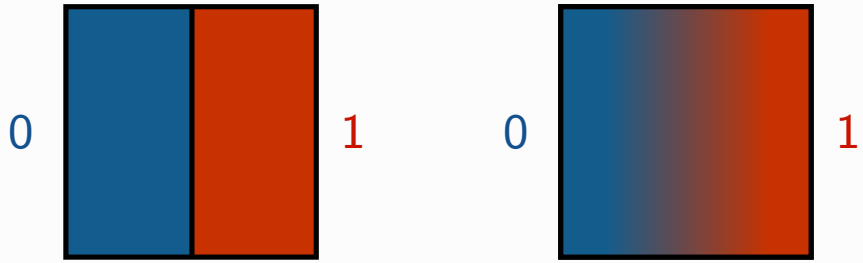
with $A_k \in \mathbb{M}^{2\times 2}_{\mathrm{skew}}$ and $\{E_k\}_{k=1}^{M}$ Caccioppoli partition of Ω. We interpret A as a linearised polycrystal.



Proof Strategy: We are minimising an anisotropic total variation functional. By Coarea formula we select the levels with minimal perimeter, definying the Caccioppoli partition.

Open Question: Are all minimisers piecewise constant? Uniqueness?

Essential: that the boundary condition is piecewise affine on the whole $\partial\Omega$.



Conclusions and Perspectives

Conclusions:

- A variational model for linearised polycrystals with infinitesimal rotations between the grains, deduced by Γ-convergence.

Conclusions and Perspectives

Conclusions:

- A variational model for linearised polycrystals with infinitesimal rotations between the grains, deduced by Γ-convergence.
- Networks of dislocations are obtained as the result of energy minimisation, under suitable boundary conditions.

Conclusions and Perspectives

Conclusions:

- A variational model for linearised polycrystals with infinitesimal rotations between the grains, deduced by Γ-convergence.
- Networks of dislocations are obtained as the result of energy minimisation, under suitable boundary conditions.

Open Questions:

- Uniqueness of piecewise constant minimisers?

Conclusions and Perspectives

Conclusions:

- A variational model for linearised polycrystals with infinitesimal rotations between the grains, deduced by Γ-convergence.
- Networks of dislocations are obtained as the result of energy minimisation, under suitable boundary conditions.

Open Questions:

- Uniqueness of piecewise constant minimisers?
- Dynamics for linearised polycrystals?

  Taylor. Bull. Amer. Math. Soc. (1978).

  Chambolle, Morini, Ponsiglione. Comm. Pure Appl. Math (2017).

Conclusions and Perspectives

Conclusions:

- A variational model for linearised polycrystals with infinitesimal rotations between the grains, deduced by Γ-convergence.
- Networks of dislocations are obtained as the result of energy minimisation, under suitable boundary conditions.

Open Questions:

- Uniqueness of piecewise constant minimisers?
- Dynamics for linearised polycrystals?
  Taylor. Bull. Amer. Math. Soc. (1978).
  Chambolle, Morini, Ponsiglione. Comm. Pure Appl. Math (2017).
- Γ-convergence analysis starting from a non-linear energy?
  Namely, considering small deformations $v = x + \varepsilon u$. Now the Burgers vectors are $\varepsilon\xi$ and the equivalent rescaling is $\varepsilon^2 N_\varepsilon |\log \varepsilon|$.
  Müller, Scardia, Zeppieri. Indiana University Mathematics Journal (2014).

Thank You!